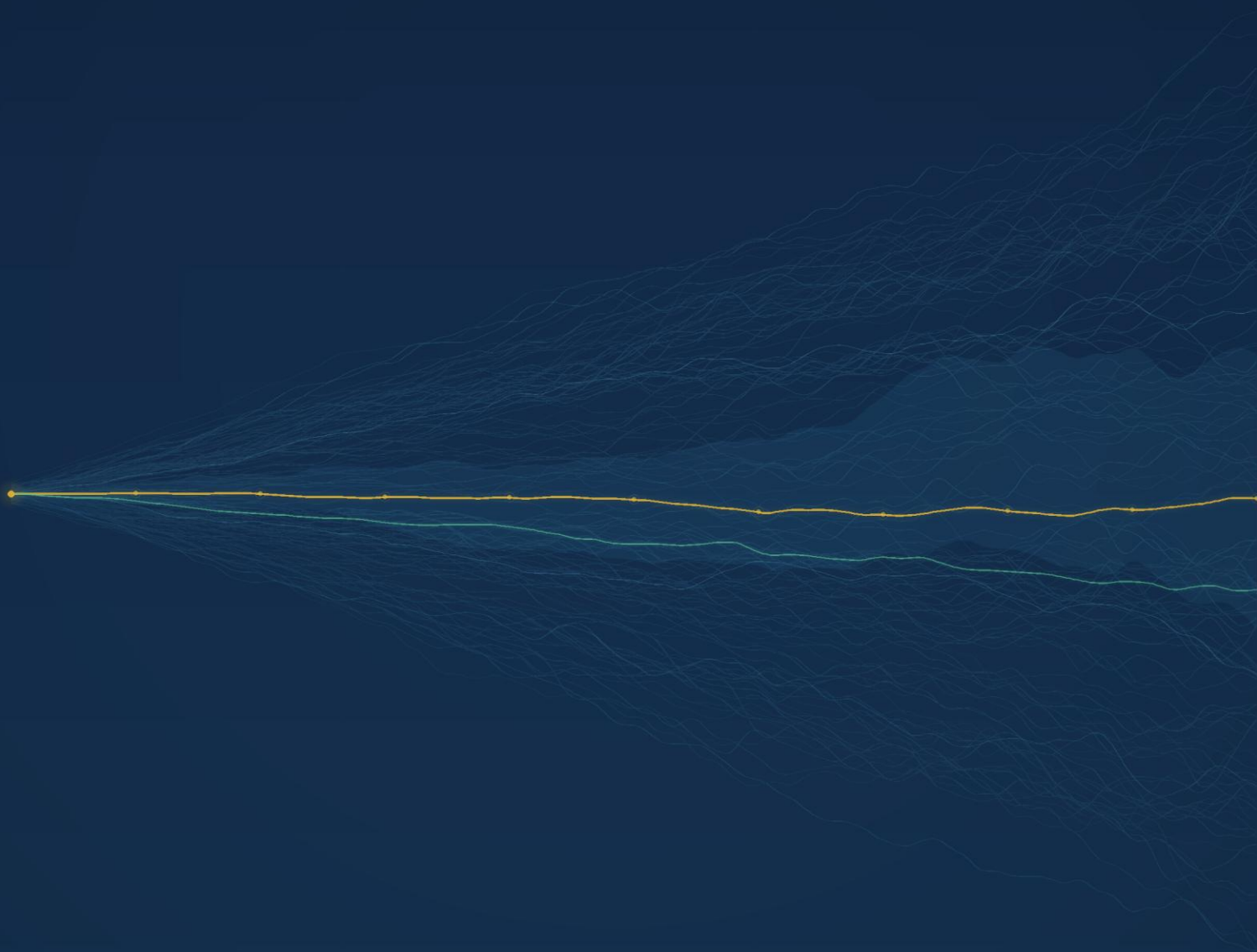


Australian Retirement Planning Tool

User Guide & Technical Reference



Version 17.2 - March 2026

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PART 1: GETTING STARTED

The calculator uses a two-column layout on desktop: inputs on the left, scenarios and results on the right. This allows simultaneous viewing of your parameters and outcomes without scrolling.

1.1 Overview

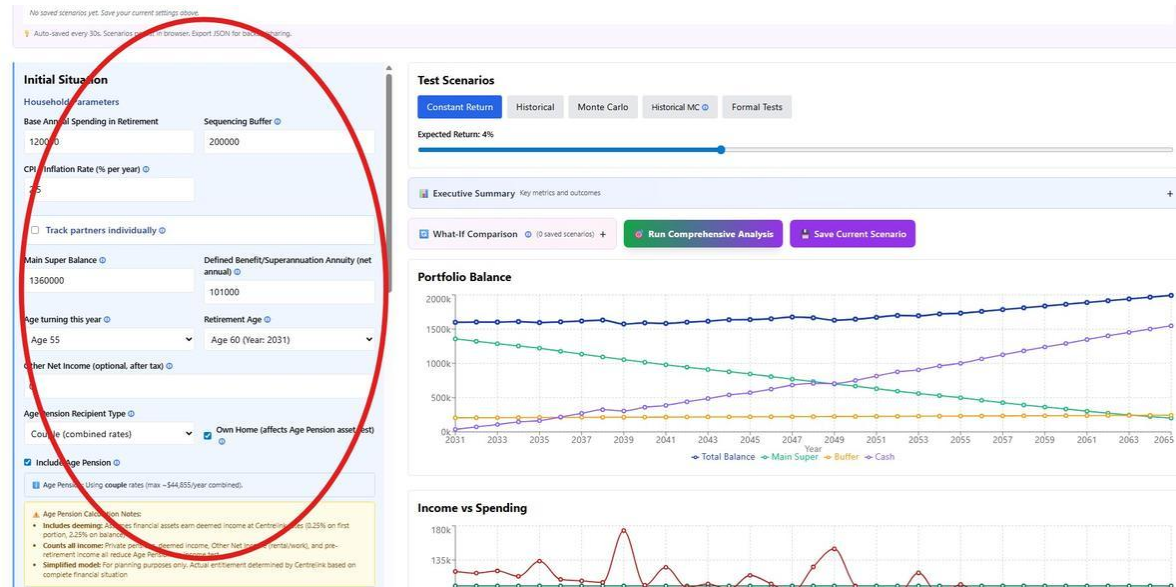
The Australian Retirement Planning Tool is a comprehensive Monte Carlo-based retirement calculator designed for Australian retirees. It models superannuation drawdowns, age pension eligibility, spending patterns, aged care costs, and death scenarios to project portfolio sustainability over retirement.

Key Features

- Monte Carlo simulation with 1,000+ scenarios and historical back testing using AU and US data (1928–2025)
- Couple tracking with individual partner management and death scenarios
- Australian Age Pension modeling with accurate asset and income tests
- Aged care cost modeling (RAD and annual fees)
- Dynamic spending guardrails with automatic adjustments
- Multiple test scenarios: constant return, historical periods, formal stress tests
- Real vs nominal dollar display modes
- Variable simulation horizon driven by Modelled Death Age (ABS life expectancy default)
- Stochastic mortality sampling in Monte Carlo — each run draws a different death age
- Longevity & Outcome Analysis chart (scatter + ruin age histogram)
- Comprehensive charts with percentile bands and executive summary dashboard

1.2 Quick Start Guide

Basic Setup (5 minutes)



1. Initial Situation

- Age turning this year: The age you will turn (or have turned) this calendar year
- Retirement Age: When you plan to retire (pension starts)
- Super Balance at Retirement: Total superannuation balance at retirement
- Sequencing Buffer: Cash for sequence risk

2. Spending

- Base Annual Spending in Retirement: Essential plus discretionary spending in retirement
- Spending Pattern: Choose constant (increasing with CPI) or declining (based on JP Morgan research)

3. Pension

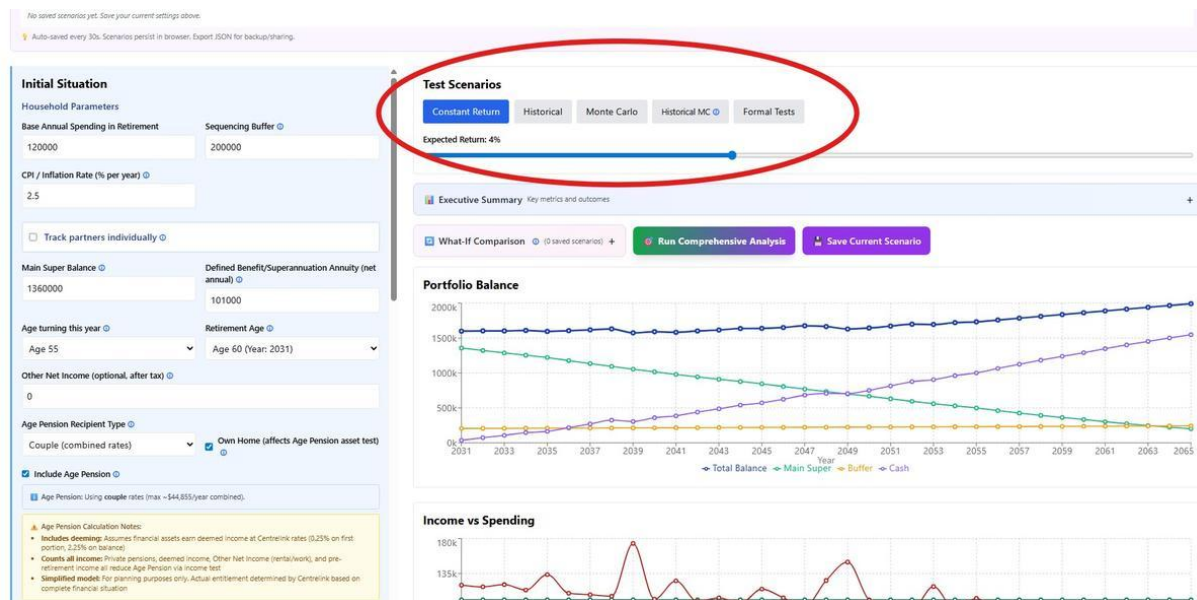
- Defined Benefit Pension: annuity/defined benefit amount (if applicable)
- Include Age Pension: Enable to model government pension (recommended)
- Pension Recipient Type: Single or Couple - affects age pension rates and asset test thresholds
- Assessable Assets Outside Super: Financial assets outside super (bank, shares, investment property equity) counted in the Age Pension means test. Enter in today's dollars.

4. Other Income (Optional)

- Other Net Income: Primary income source like part-time work, rental income, dividends (if applicable)
- Start Year: When it begins (default 1 = immediate)
- Duration: How many years it lasts (blank = indefinite)
- Index to CPI: When enabled increases the net income by inflation
- Employment Income – Work Bonus Eligible: Select if the income is from employment or self-employment.
- Additional Income Streams: Click “Add Income Stream” for extra sources (e.g., overseas pension, second rental). Each has its own amount, start year, duration, and CPI indexing toggle.

5. Run Simulation

- Select test scenario: Start with 'Constant Return' at 5%



- Review charts: *Portfolio Balance* and *Income vs Spending*
- Check *Executive Summary*: Expand to see success/fail status and key metrics

Understanding Results

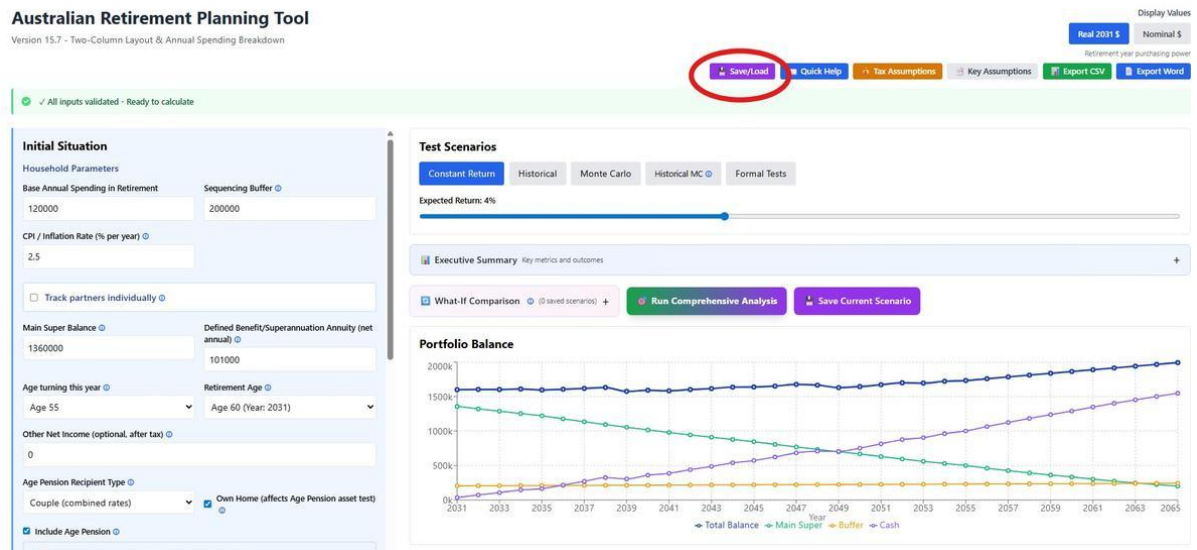
Success = Portfolio lasts to the Modelled Death Age with balance $\geq$ \$0

Failure = Portfolio depleted before the Modelled Death Age

After reviewing basic results, explore Monte Carlo simulation for uncertainty analysis or formal stress tests to evaluate resilience.

1.3 Save/Load Scenarios

The Save/Load feature allows you to save multiple retirement scenarios, compare different strategies, and backup your planning work. All scenarios are stored locally in your browser with optional JSON export for permanent backup.



Saving a Scenario

1. Configure all calculator settings (ages, super balance, spending, test scenarios, etc.)
2. Click the "Save/Load" button at the top of the page
3. Enter a descriptive scenario name (e.g., "Base Case", "Conservative 6% Returns", "Retire at 65")
4. Click "Save"
5. Your scenario is saved to browser storage and appears in the "Saved Scenarios" list

Saved scenarios persist across browser sessions and include all calculator settings: financial inputs, test scenario selection, Monte Carlo settings, couple tracking, aged care, debts, and display preferences.

Loading a Scenario

1. Click "Save/Load" button
2. Find your scenario in the "Saved Scenarios" list
3. Click the "Load" button next to the scenario name
4. All settings are instantly restored
5. The Scenario Manager closes automatically

Note: Loading a scenario overwrites your current unsaved work.

Export/Import (JSON Backup)

Export to JSON: Creates a downloadable .json file containing your scenario. Use this for:

- Permanent backup (not tied to browser)
- Sharing scenarios with financial advisors or family

- Transferring between computers/browsers
- Version control (save multiple versions with dates)

Export to JSON: Saves a scenario as a JSON file.

1. Click "  Save/Load"
2. Click "  Export"
3. Enter a descriptive scenario name
4. File is saved as a downloadable JSON file (check your download history)

Import from JSON: Restores a previously exported scenario.

1. Click "  Save/Load"
2. Click "  Import"
3. Select your .json file
4. Settings load immediately

JSON files are human-readable text files containing all scenario data. They're small (typically 5-15 KB) and can be stored anywhere.

Auto-Save

The calculator automatically saves your inputs to your browser's local storage every time you make a change. If you close your browser and return later, your inputs will be restored.

There is no expiry on auto-saved inputs. Your settings will persist until you clear your browser's local storage, use a different browser or device, or explicitly load a different saved scenario.

Note that auto-save preserves your input settings only — it does not save Monte Carlo or Historical MC results, which must be re-run after each session

Best Practices

Naming Scenarios: Use descriptive names that explain the key differences:

- "Base Case 2026" - Your most realistic scenario
- "Conservative 6% Returns" - Lower return assumption
- "Retire 2030 Age 60" - Specific retirement timing
- "With Aged Care" - Including aged care costs

Avoid generic names like "Scenario 1" or "Test"

Backup Strategy:

- Save important scenarios in browser (quick access)
- Export to JSON monthly (permanent backup)
- Store JSON files in cloud storage (Google Drive, Dropbox)
- Name files descriptively: "Retirement_BaseCase_2026-02.json"

Browser Storage: Scenarios are stored in the browser local storage (5-10MB limit). You can

typically store 100-500 scenarios. Export old scenarios to JSON and delete from your browser if you approach the limits.

1.4 Understanding the Interface

Display Mode Toggle

Choose between Real and Nominal dollar display in the top-right corner:

Real \$ (Current Year): Values adjusted to current year purchasing power. Example: a nominal balance of \$133,469 in Year 10 shows as \$100,000 — reflecting that \$133,469 in 10 years has the same buying power as \$100,000 today, assuming 2.5% CPI. Use this to understand future amounts relative to your current cost of living.

Real \$ (Retirement Year): Values adjusted to retirement year purchasing power. Example: a nominal balance of \$133,469 in Year 10 shows as \$104,622 — deflated back to retirement-year dollars using 10 years of 2.5% CPI. Use this to track how your spending and balances evolve in real terms relative to when you retired.

Nominal \$: Future dollar amounts without inflation adjustment. Example: \$133,469 in Year 10 shows as \$133,469. Use this to match bank statements, fund projections, and Centrelink estimates..

Australian Retirement Planning Tool
Version 17.1 - Variable Horizon & Longevity Analysis

Display Values
Real 2026 \$ Real 2031 \$ Nominal \$

Save/Load Quick Help Tax Assumptions Key Assumptions Export CSV Export W...

✓ All inputs validated — ready to calculate

Sex: Male Female
Modelled Death Age: 100
40 years of retirement
Other Net Income (optional, after tax): \$20,000
Duration (years, optional): Indefinite
Additional Income Streams: Add Income Stream
Pension Recipient Type: Single
Include Age Pension: [checked] Own Home: [checked]
Accessible Assets Outside Super (\$): 0
Non-Portfolio Income Summary
Retirement Spending

Scenario Selection
Constant Return Monte Carlo Historical MC Stress Tests Historical Period

Historical Monte Carlo
Historical Data Series: AU Balanced AU Equities AU Bonds US S&P 500
60% ASX All Ords Accumulation + 40% Australian Composite, best proxy for a typical Australian industry super balanced fund.
Simulations: 1,000
Block Size: 5
Sampling Method: Block bootstrap
Shuffled Years: 100
25-Year Blocks: Full consecutive periods
Run Historical MC
Success: 100.0% (overlapping) 1,000 runs

Executive Summary Key metrics and outcomes
Longevity & Outcome Analysis
What-If Comparison
Age Pension Calculation Details Year-by-year breakdown of Age Pension means testing
Historical MC: Median from historical sampling, Success: 100.0%
Portfolio Balance - Historical Median
6000%

Collapsible Sections

Most input sections can be expanded/collapsed using +/- buttons. This keeps the interface clean while allowing deep configuration when needed.

1.5 Enhanced Warning Systems

Early Portfolio Depletion Warning

If your portfolio depletes within the first 5 years, a red warning banner appears above the Portfolio Balance chart showing which year the depletion occurred. The banner suggests four actions to address the shortfall: increase your expected return to better reflect your asset allocation, reduce annual spending, increase the sequencing buffer, or run Monte Carlo for a probabilistic view of outcomes.

A separate warning banner appears in couple tracking mode if one partner has not yet retired, leaving insufficient accessible funds to cover expenses. The fix is to add the working partner's salary as pre-retirement income, or to align the retirement ages.

Stale Results Detection (Monte Carlo)

Monte Carlo and Historical Monte Carlo simulations can take several seconds to run. When you first switch to Monte Carlo mode, the simulation runs automatically. After that, it does not re-run when you change settings — you must click the Run button manually.

When settings change after a run, the calculator provides visual stale detection: the Run button turns orange and pulses, the button text changes to " Re-Run Monte Carlo (Settings Changed)", and a warning appears below the button. Charts continue showing the previous results until you re-run. Once you click Run, the button returns to its normal colour and the warning disappears.

Data Currency Notice

A small indicator appears at the top of the calculator showing the currency status of the financial parameters built into the tool — Age Pension rates, deeming rates, Work Bonus amounts, super contribution caps, and similar figures that are updated periodically by government. When all data is current the indicator is green and shows "All data current." When a parameter is due for update within 30 days it turns amber; when it is overdue it turns red. Click the indicator to expand a table showing each data item, its next scheduled review date, and a brief note on the source. This is particularly relevant around the Age Pension indexation dates of 20 March and 20 September each year, when rates and thresholds are updated by Services Australia.

PART 2: CORE INPUTS

2.1 Initial Financial Situation

Age Turning This Year & Retirement Age

Age Turning This Year determines life expectancy calculations. Retirement Age marks Year 1 of the simulation when superannuation access begins and private pensions commence. For couples, Year 1 starts when the first partner retires (see Section 2.5). The simulation runs to your Modelled Death Age (see below), so the duration depends on both your retirement age and the death age you set. The formal stress test D1 (Extreme Longevity) overrides this with a fixed 45-year horizon to age 105.

Modelled Death Age & Sex

Sex: Select Male or Female. This sets the default Modelled Death Age from the ABS 2020–2022 period life tables and is used for stochastic mortality sampling in Monte Carlo (see below).

Modelled Death Age: The age at which the simulation ends. Defaults to the ABS life expectancy for your sex and current age (for example, age 84 for a 60-year-old male). You can override this manually. You can reset the age by selecting *Reset to ABS default*. The field shows the resulting number of retirement years below the input (e.g. “30 years of retirement”). Setting a longer death age extends the simulation horizon and will reduce Monte Carlo success rates if spending outpaces income over the longer period.

Variable death age in MC: When enabled (checkbox at the bottom of the Death Scenario Modelling tab), each parametric Monte Carlo run samples a different death age from a normal distribution centred on the ABS life expectancy for your sex and current age, with a standard deviation of 10 years. This means some runs end at age 74, others at 102, reflecting genuine longevity uncertainty. The deterministic simulation always uses the fixed Modelled Death Age you set. This option is only available for parametric Monte Carlo — Historical Monte Carlo uses fixed 35-year windows to preserve the integrity of historical return sequences.

Portfolio Buckets

Super Balance at Retirement / Main Super Balance: Your projected superannuation balance at the date of retirement. If retiring soon, use your current balance. If years away, use an online super projection calculator or your fund’s estimate to determine what it will be when you retire¹. Super is subject to minimum drawdown rules (4-14% based on age). Returns are earned based on the selected scenario. This account is drawn down in accordance with the withdrawal waterfall (see Section 7.2).

Sequencing Buffer: Conservative allocation (cash) earning a configurable return (default 3% per year). Provides protection against sequence-of-returns risk by funding spending during market downturns. A typical allocation is one to two years of spending. The return rate is set in the *Cash / Buffer Return Rate* field, alongside the CPI rate.

Cash Account: Emergency fund earning the same configurable return as the Sequencing Buffer (default 3% per year). Accessed first in the withdrawal hierarchy. Holds overflow cash when pension income plus minimum super drawdowns exceed expenses.

Homeownership Status

Affects Age Pension asset test thresholds. Homeowners have a lower asset test threshold (\$321,500 for singles) because the principal residence is exempt. Non-homeowners have a higher threshold (\$579,500 for singles) to compensate for housing costs. Select 'Yes' if you will own your home at retirement.

¹ Most Australian super calculators project an estimate of your retirement balance in today’s dollars. This is to show what your money will be worth at retirement in today’s dollars, rather than what your actual account balance will be. If this is the case, select 2026\$ as the input, and the calculator will convert it to retirement year dollars.

2.2 Retirement Spending

Base Annual Spending in Retirement

Essential plus discretionary expenses in retirement. Automatically adjusted each year in accordance with the selected spending pattern (Constant Real or JP Morgan Declining). Includes groceries, utilities, transport, entertainment, travel, insurance, and rates. Excludes mortgage payments (use the Debt Repayment section), aged care costs (modelled separately), and one-off or splurge spending.

If retirement is several years away, switch the input toggle to today's dollars and enter your current spending — the calculator will convert it to retirement-year dollars automatically using your CPI rate and years to retirement.

A "Not sure? See spending benchmarks" link appears below this field showing reference ranges for homeowners (December 2025 figures): Comfortable ~\$77,375/\$54,840 couple/single, Modest ~\$51,299/\$35,503 couple/single, and Above Average \$150k–\$200k for couples (BDO). A link to the Moneysmart Budget Planner is also provided for building a personalised estimate. Note that these benchmarks assume homeownership — renters will need more.

Find Sustainable Spending

A "Find sustainable spending" link appears below the spending benchmarks. Clicking it expands a panel that solves for the maximum base spending your portfolio can sustain, using your current inputs across the full simulation engine — Age Pension means testing, couple tracking, guardrails, pension income, and all other settings are included.

Constant Return Mode: Uses a binary search to find the base spending level that leaves your portfolio at the target ending balance at your Modelled Death Age, under the current constant return assumption. Enter a target ending balance (default \$0 for full spend-down, or a higher figure as a bequest target). The result shows sustainable base spending in retirement-year dollars, the initial withdrawal rate, ending balance, and simulation horizon. A green Apply button populates the Base Annual Spending field directly.

Monte Carlo Mode: Finds the maximum base spending where your Monte Carlo success rate meets a chosen threshold (70%, 75%, 80%, 85%, or 90%). Each iteration of the search runs approximately 250 full Monte Carlo simulations using your current expected return and volatility settings. The solve typically takes several seconds — allow up to 30 seconds on older devices. The result shows the sustainable spending figure, the achieved success rate, and a sensitivity strip showing the success rate at \$5,000 above and below the answer so you can see how steeply the success rate changes if you adjust spending slightly. This is the more rigorous of the two modes: the constant-return solve gives a deterministic ceiling, while the MC solve gives a probabilistically grounded answer accounting for sequence-of-returns risk.

Splurge spending interaction: The sustainable base spending figure returned by both modes already accounts for your configured splurge spending. The binary search includes splurge on top

of the base figure in every iteration, so applying the result to the spending field will sustain both the base spending and the splurge spending simultaneously.

Spending Patterns (see 3.4 Withdrawal Strategy)

Constant Real: Constant spending adjusted for inflation only. This is a conservative assumption.

JP Morgan (Declining): Spending curve declines gradually with age (relative to CPI) based on Blanchett (2014) and JP Morgan (2024) research adjusted for Australia. This models three phases. In the early "go-go" years (years 1–10), real spending declines ~1.0% per year as initial retirement enthusiasm moderates. In the "slow-go" years (years 11–20), the decline accelerates to ~1.5% per year as activity reduces. In the "no-go" years (21+), spending tapers to ~0.5% annual real decline. Note that aged care costs (if enabled) are modelled separately on top of this curve, which can cause total spending to rise again in later years. This is the default and is generally more realistic than CPI (Level).

Splurge Spending

Additional spending for major purchases or lifestyle upgrades. Configure start age, duration, annual amount, and ramp-down period. Example: \$30k/year for 10 years starting at 65 for international travel, ramping down over final 3 years.

2.3 Pension Income

Private Pension (Annuity/Defined Benefit)

Enter your annual **defined benefit pension or superannuation annuity income** (after tax) that will start at retirement. This includes:

- **Public sector pensions:** PSS, CSS, DFRDB
- **Commercial lifetime annuities:** From super or investment
- **Other defined benefit schemes:** Any guaranteed lifetime income

The calculator automatically indexes this income to inflation (CPI) each year. Enter the **net annual amount** (after tax) you expect to receive starting from your retirement year. If you have an annuity or pension that is not linked to CPI, you can use the Other Net Income feature below.

Age Pension

Government payment for eligible retirees aged 67+. Eligibility determined by:

Asset Test: Total assessable assets excluding principal residence. Homeowner threshold: \$321,500 (single), \$481,500 (couple). Pension reduces \$3 per fortnight for every \$1,000 above threshold. Full cutoff: \$722,000 (single), \$1,085,000 (couple).

Income Test: Annual income including deemed investment income. Free area: \$5,668/year (single), \$9,880/year (couple combined). Pension reduces 50c per dollar above threshold.

Maximum Rates (2025): \$31,224/year (single), \$47,071/year (couple combined)

Calculator applies the lower result of asset or income test. For couples with different ages, when only one partner has reached 67 the eligible partner receives the member-of-couple rate

(approximately \$23,536/yr). When both reach 67, the full couple rate applies (~\$47,071/yr combined). After one partner dies, the survivor receives the full single rate (\$31,224/yr).

2.4 Other Net Income

Model additional income sources during retirement beyond pensions and Age Pension. Examples include part-time work, rental properties, investment income, trust distributions, or consulting. Income can start at any year and run for a specified duration or indefinitely.

Configuration Fields

Other Net Income (after tax): Income per year after tax. Examples: \$30,000 for part-time work, \$25,000 for rental income, \$15,000 for trust distributions.

Start Year: Which year of retirement this income begins (1-35). Year 1 = first year of retirement. Default is 1 (starts immediately). Example: Enter 5 if income starts in the 5th year of retirement.

Duration: How many years the income lasts (1-35 years). Leave blank for indefinite income that continues throughout retirement. Examples: 10 years for part-time work, 15 years for rental property, blank for ongoing trust income.

Index to CPI: When enabled (default), income increases with inflation each year. Disable for fixed contracts or annuities that don't adjust for inflation. Examples of indexed income: part-time work wages, rental properties. Examples of fixed income: fixed annuity contracts, non-indexed leases, fixed bond income.

Common Scenarios

Part-Time Work (Indexed): Amount \$30,000, Start Year 1, Duration 10, CPI Indexed ☒ → Work part-time for first 10 years, wages increase with inflation (Year 1: \$30,000, Year 10: ~\$37,500 at 2.5% CPI)

Rental Property: Amount \$25,000, Start Year 1, Duration 15 → Rental income until property sold in Year 15

Delayed Consulting: Amount \$40,000, Start Year 3, Duration 5 → Take 2 years off, then consult for 5 years

Trust Distributions: Amount \$20,000, Start Year 1, Duration (blank) → Ongoing income throughout retirement

Partner's Work (Couple Mode): Amount \$25,000, Start Year 5, Duration 8 → Second partner works for 8 years after retiring 4 years later

Fixed Annuity: Amount \$25,000, Start Year 1, Duration (blank), CPI Indexed ☐ → Lifetime fixed annuity that doesn't adjust for inflation, stays at \$25,000/year throughout retirement

Timing and Indexing

Income streams use a year-based system where Year 1 is the first year of retirement — or, in couple mode, the year the first partner retires. All years are relative to this start point, so if Partner 2 retires 5 years later, their income stream starting in Year 6 begins in their first year of retirement.

By default, income amounts increase with inflation each year. Untick the "Index to CPI" checkbox for income that is fixed in dollar terms, such as a fixed annuity or a non-indexed lease.

Other Income appears as an orange stacked area in the Income vs Spending chart, making it easy to see exactly when it starts and stops. It is also included in What-If Comparison tables, CSV exports, and the Executive Summary.

Important: Other Net Income is counted in the Centrelink income test and will reduce your Age Pension entitlement. The calculator models this automatically, but you should verify your Age Pension estimate directly with Centrelink if you have rental income, part-time work, or other assessable income sources, as your actual circumstances may differ from the calculator's assumptions.

Work Bonus Eligibility: When Age Pension is enabled, an "Employment income — Work Bonus eligible" checkbox appears beneath the main Other Net Income field, and at the bottom of each Additional Income Stream card. Tick this checkbox for income that qualifies as employment or self-employment income (wages, salary, ABN consulting). Do not tick it for rental income, dividends, defined benefit pensions, annuities, or overseas pensions — those sources are not eligible for the Work Bonus and remain fully assessable.

Centrelink exempts up to \$300/fortnight (\$7,800/year) of employment income from the Age Pension income test for eligible pensioners aged 67+. Unused entitlement accumulates in a "bank" up to a cap of \$11,800 per person. The calculator models this accrual year by year: if employment income is low in early years, the bank grows and can offset higher earnings in later years. For couples, each partner has a separate bank. Both the annual accrual (\$7,800) and the cap (\$11,800) are CPI-indexed within the simulation. The bank starts at \$0 (conservative — no prior balance assumed).

2.5 Couple Tracking

Couple tracking enables individual modelling of each partner's superannuation, pension income, retirement timing, and death scenarios. When enabled, the calculator tracks each partner separately and handles the transition to survivor mode automatically when a death scenario is active.

Warnings (1)

- Partner 2 not yet retired but has no pre-retirement income

These warnings highlight unusual values. Review them to ensure they're intentional.

Initial Situation

Household Parameters

Base Annual Spending in Retirement: 138500

Sequencing Buffer: 125000

CPI / Inflation Rate (% per year): 2.5

☒ Track partners individually

Individual Partner Tracking

Partner 1		Partner 2	
Name	Tim	Name	Simone
Age turning this year	56	Age turning this year	56
Gender	Male	Gender	Female
Super Balance	1300000	Super Balance	0
Retirement Age	60	Retirement Age	60
Defined Benefit/Annuity (net annual)	8049	Defined Benefit/Annuity (net annual)	0
Pre-Retirement Income (net annual)	90000	Pre-Retirement Income (net annual)	0
Reversionary Rate (%)	67	Reversionary Rate (%)	67
Death Age (for scenario modeling)		Death Age (for scenario modeling)	

Test Scenarios

Constant Return | Historical | Monte Carlo | Historical MC | Formal Tests

Expected Return: 7.5%

Executive Summary Key metrics and outcomes

What-If Comparison (0 saved scenarios) +

Run Comprehensive Analysis **Save Current Scenario**

Portfolio Balance

Y-axis: 0k to 3200k. X-axis: Year (2030 to 2064). Legend: Total Balance, Main Super, Buffer, Cash.

Income vs Spending

Y-axis: 100k to 200k. X-axis: Year (2030 to 2064). Legend: Income, Spending.

Partner Configuration

Note: The "Track partners individually" toggle only appears when the Pension Recipient Type is set to Couple. You will find the Pension Recipient Type selector in the Age Pension Settings section, just above the Include Age Pension checkbox. Set it to Couple first, then the couple tracking toggle will appear in the Initial Situation section.

Each partner is configured with the following fields:

Name — used to identify each partner in the charts and partner summary panels.

Age Turning This Year — the age each partner turns this calendar year. Used to calculate years to retirement and to set the default death age from ABS life expectancy tables.

Retirement Age — when each partner retires. This is when their superannuation becomes accessible and their pre-retirement income ceases. Year 1 of the simulation starts when the first partner reaches their retirement age.

Super Balance — each partner's projected superannuation balance at their individual retirement date.

Pension Income — annual defined benefit or annuity income (after tax) starting from retirement. Automatically indexed to CPI. Examples include PSS, CSS, DFRDB, or commercial lifetime annuities.

Reversionary Rate — the percentage of pension income that continues to the surviving partner after death. Typically 67% for PSS and similar defined benefit schemes. Enter 0 if no reversionary benefit applies.

Sex — used to set the default death age from ABS 2020–2022 life expectancy tables. The death age updates automatically when sex is changed, unless you have manually overridden it.

Death Age — the age used in death scenario modelling. Has no effect when Both Alive is selected; only activates when Partner 1 Dies or Partner 2 Dies is chosen.

Pre-Retirement Income — the working partner's salary or other income while the other partner has already retired. This prevents the "insufficient funds" warning that can occur during the gap between staggered retirement dates, and is counted in the Age Pension income test.

Year 1 Anchoring

Critical Concept: Year 1 starts when the first partner retires (earliest retirement age). Partners retiring later continue earning pre-retirement income until their retirement. Pensions start at each partner's individual retirement age.

Example:

Partner 1: Current age 55, retires at 60

Partner 2: Current age 56, retires at 65

Year 1: Partner 1 is 60 (retired), Partner 2 is 61 (working, earning pre-retirement income)

Year 6: Partner 1 is 65, Partner 2 is 66 (just retired, pension starts)

Super Withdrawal Logic

Each partner's super is tracked separately. Withdrawals are proportional to accessible balances:

Example:

Partner 1 super: \$800,000 (retired - accessible)

Partner 2 super: \$200,000 (retired - accessible)

Total accessible: \$1,000,000

Need to withdraw: \$50,000

→ Partner 1 withdraws: $\$50,000 \times 80\% = \$40,000$

→ Partner 2 withdraws: $\$50,000 \times 20\% = \$10,000$

Important: Pre-retirement partners cannot access their super. Only retired partners' balances are accessible.

Simulation Length

Simulation Length (years): How many years to run the simulation from the first partner's retirement. This is the primary control for how long the couple scenario runs and is completely independent of the partner death ages configured below. Partner death ages only affect death scenario modelling — they have no bearing on how long the simulation runs. The field displays the equivalent end age for the first retiree as a reference. Default is derived from the longer-lived partner's ABS life expectancy at first load, and can be adjusted freely.

Death Scenario Modeling

This feature lets you explore "what if one of us dies early?" as a planning scenario. It does **not** model life expectancy or mortality probability — it's a manual toggle to stress-test your plan.

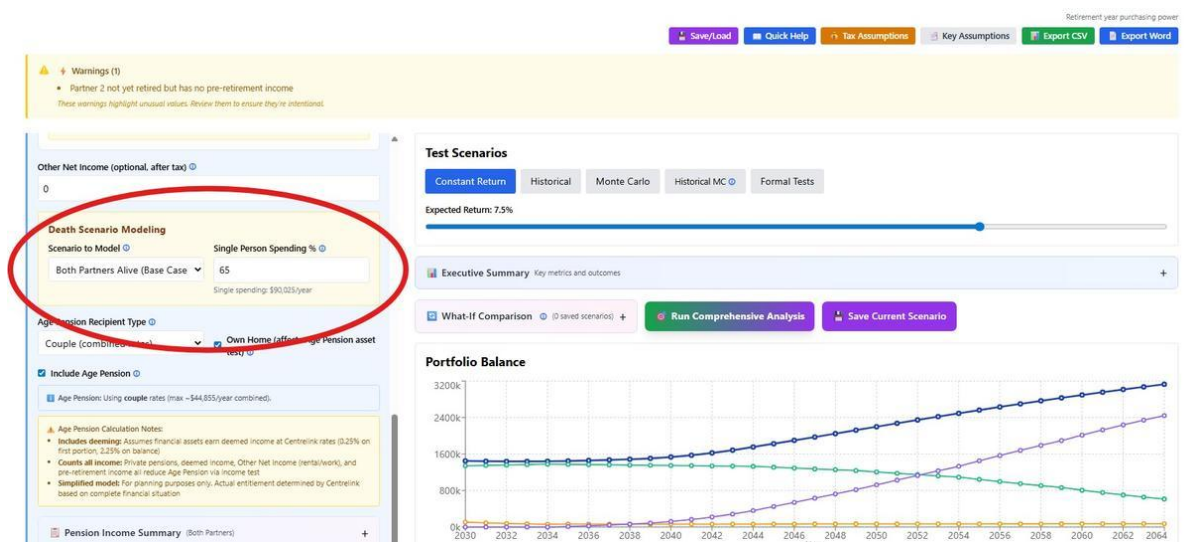
Important: "Both Alive" does not mean "both die at their death ages." It means both partners are assumed to be alive for the entire simulation period. The death ages you set have no effect in this mode — they only activate when you select a death scenario.

Three scenarios:

- **Both Alive (default):** Both partners are alive throughout the entire simulation. Couple spending, couple pension rates, and both income streams continue for the full period. Death ages are ignored.
- **Partner 1 Dies (at specified death age):** Models the financial impact of Partner 1 dying at their set death age. At that point: superannuation balance transfers to Partner 2, defined benefit/annuity pension reduces to the reversionary rate, household spending drops to the single-person rate (default 70% of couple spending), and age pension switches from couple to single rate. Partner 2 continues as sole survivor for the remainder of the simulation.
- **Partner 2 Dies:** Same logic, with Partner 2 dying and Partner 1 as survivor.

Why it works this way: Modelling both partners dying at specific ages would simply truncate the simulation with no remaining beneficiary. The key planning question is whether the surviving partner can sustain themselves financially if one partner dies first. These scenarios address that question.

Variable death age in MC: A checkbox at the bottom of the Death Scenario Modelling panel enables stochastic mortality for parametric Monte Carlo. When enabled, each MC run independently samples a death age for each partner from a normal distribution centred on their ABS life expectancy (SD = 10 years). The simulation horizon for each run extends to the longer-lived partner's sampled age, so run lengths vary across the MC ensemble — producing a more realistic spread of outcomes than a fixed horizon. When this option is off, all runs use the fixed Simulation Length set above. It has no effect on the deterministic simulation.



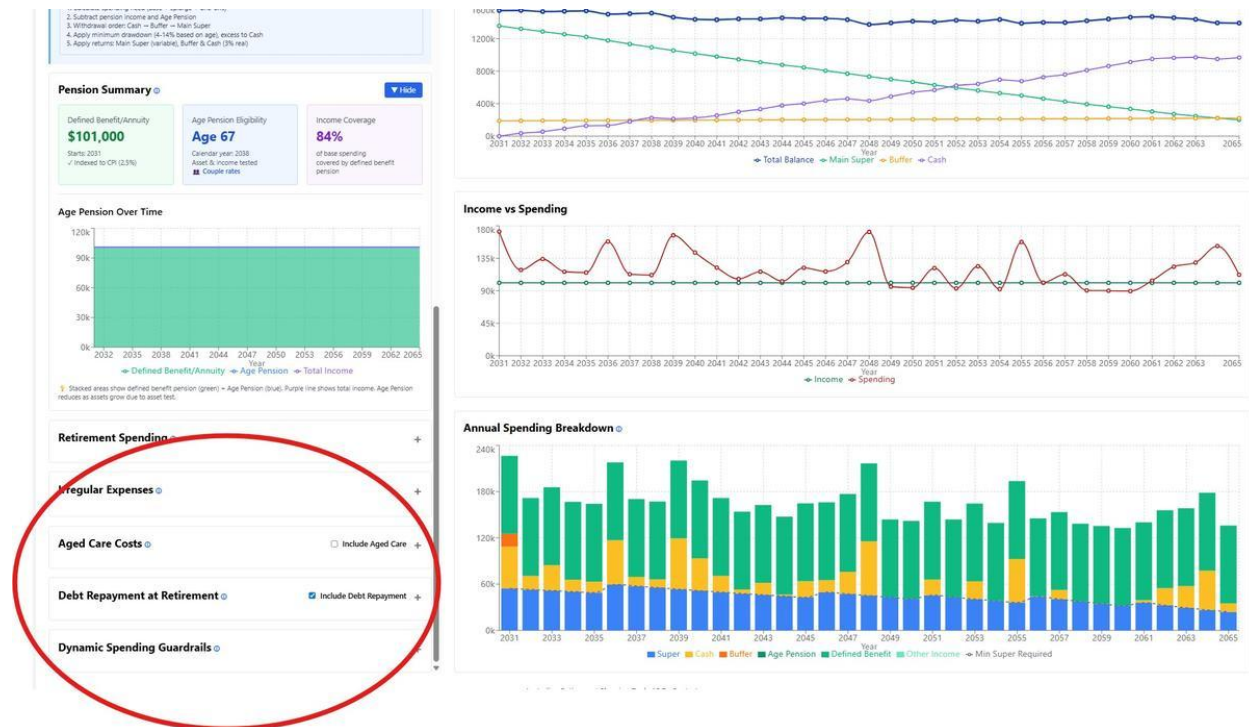
Age Pension for Couples

Accurately models Centrelink rules based on individual eligibility:

- Both under 67: No age pension
- One 67+, one under: Member-of-couple rate (~\$23,536/year) for eligible partner only — not the full single rate
- Both 67+: Couple rate (~\$47,071/year combined), split 50/50 in individual charts
- After death: Survivor receives single rate

Asset and income test thresholds automatically adjust based on whether single or couple rates apply.

PART 3: ADVANCED FEATURES



3.1 Withdrawal Strategy

The Withdrawal Strategy section controls how much you draw from your portfolio each year. It has two independent components that can be combined in any combination: a base spending pattern and up to two modifiers. Australian superannuation minimum drawdown requirements apply regardless of strategy — they act as a hard floor that the ATO mandates independently of any spending decision you make.

Base Spending Pattern

The spending pattern sets the shape of real (inflation-adjusted) spending over your retirement. It does not change the total amount you tell the calculator you want to spend — it determines how that spending evolves year by year.

Constant Real

Your base spending is increased by CPI each year so that the real (purchasing-power-adjusted) amount stays constant throughout retirement. If you set \$120,000 as your base spending, you will always have the equivalent of \$120,000 in today's purchasing power, regardless of how many years have passed.

Advantages: Simple and predictable. Easy to budget around. Conservative — tends to overstate late-retirement spending since most retirees naturally spend less as they age. Good for stress-testing portfolio longevity at the upper bound of spending.

Disadvantages: Not representative of how retirees actually spend. Research consistently shows real spending declines with age. Using Constant Real often results in the simulation showing large unspent balances at death — technically a “successful” plan but one that undersells what you could have spent.

When to use: Conservative stress-testing; when you genuinely believe your spending will not decline with age (e.g. significant ongoing care costs); as an upper-bound comparison against the J.P. Morgan Curve.

J.P. Morgan Curve

Real spending declines gradually with age, following three phases observed in actual retiree expenditure data (Blanchett 2014; J.P. Morgan 2024), adjusted for Australia:

- Go-Go years (years 1–10): Real spending declines ~1.0% per year. Active early retirement — travel, activities, dining out — moderates slightly as the initial burst of enthusiasm settles.
- Slow-Go years (years 11–20): Real spending declines ~1.5% per year. Activity reduces as mobility and energy decrease. Discretionary spending falls noticeably.
- No-Go years (years 21+): Real spending tapers to ~0.5% annual decline. Non-discretionary costs (health, housing) dominate; total spending stabilises at a lower level. Note: if aged care is enabled, care costs are added on top of this curve and may cause total spending to rise again in later years.

By year 20, real spending is approximately 15% below the year-1 level. By year 35 it is approximately 21% below. The nominal spending amount still increases each year due to CPI, but more slowly than inflation, so real purchasing power gradually declines as intended.

Advantages: More realistic than Constant Real for most retirees. Produces higher sustainable initial spending (you can afford more in the go-go years because you know spending will decline). Better matches the “smile-shaped” spending pattern observed in large-scale retiree surveys.

Disadvantages: The declining-spending assumption may not hold for everyone. If you have significant fixed costs that do not decline with age (ongoing care, fixed annuity commitments, support for adult children), the curve will understate your actual spending in later years.

When to use: Most retirement planning scenarios. This is the default and produces the most realistic projection for the majority of retirees. Run both patterns as a sensitivity check — the gap between them shows the value of declining spending.

Modifier 1: Forgo Inflation Adjustment

When enabled, the calculator skips the annual CPI increase in any year that follows a negative portfolio return. Nominal spending stays flat in that year rather than rising with inflation. Because the real spending base is not increased, the effect is a small real spending cut — you are spending slightly less in purchasing-power terms than you would have otherwise. In subsequent years with

positive returns, normal CPI increases resume, but the base from which they are applied remains permanently lower. The cuts are small individually but cumulative over time.

This is based on the T. Rowe Price method. Morningstar's 2025 research found it lifts the safe withdrawal rate from 3.9% to 4.3% for a 40% equity / 60% bond portfolio over a 30-year horizon — one of the best longevity improvements per unit of lifestyle sacrifice of any strategy tested.

Example: Base spending \$120,000, CPI 2.5%, Constant Real active. Year 3 sees a portfolio loss. Without this modifier, year 4 spending would be $\$120,000 \times 1.025 \approx \$123,000$. With the modifier, year 4 spending stays at \$120,000. If year 5 has a positive return, year 5 spending becomes $\$120,000 \times 1.025 \approx \$123,000$ — the same as year 4 would have been without the modifier, but one year later and \$3,000 lower than it would otherwise have been from this point forward.

Example: Base spending \$120,000, CPI 2.5%, J.P. Morgan Curve active.

Without any modifier (J.P. Morgan only):

- Year 1: \$120,000 (nominal)
- Year 5: ~\$117,900 real spending (J.P. Morgan has applied $4 \times 1\%$ real decline), but nominal is ~\$133,000 after CPI
- Year 15: ~\$108,000 real (cumulative decline through go-go and into slow-go years), nominal ~\$163,000

With Forgo Inflation added, assuming a bad year in year 3:

- Year 3 sees a negative return
- Year 4: J.P. Morgan curve still applies (real spending continues its programmed decline), but the CPI uplift is skipped. Nominal spending stays flat at approximately the year 3 level instead of rising ~2.5%
- Year 5 onwards (positive returns resume): Normal CPI increases resume from the year 4 base, which is now permanently ~\$3,000 lower than it would have been
- The J.P. Morgan real decline continues on top of this lower base

The key point: The two mechanisms operate on different dimensions. J.P. Morgan adjusts *real* spending downward according to a predetermined curve. Forgo Inflation withholds the *nominal* CPI uplift after bad years. In combination, a loss year produces both effects simultaneously — the J.P. Morgan multiplier reduces real spending by ~1%, and the foregone CPI increment means nominal spending also stays flat rather than rising 2.5%. The result is a slightly larger real spending cut in loss years than either mechanism alone would produce, which is actually the intended behaviour — loss years call for more restraint, not less.

Advantages: Very simple rule — easy to implement and explain. Low cash flow volatility; the adjustment in any single year is only the CPI increment (typically 2–3%). Works naturally with

either spending pattern. Provides meaningful longevity improvement with minimal lifestyle disruption. No configuration required.

Disadvantages: The cut is permanent — the foregone CPI increment never comes back. After several consecutive loss years, the cumulative effect can be meaningful. The strategy reacts to last year's return, which may not reflect your current portfolio position (e.g. the portfolio may have recovered intra-year). It also does not provide upside — there is no mechanism to increase spending above the CPI-adjusted level when markets perform well.

When to use: An excellent default modifier for most retirees who want some longevity protection without complex rules or large year-to-year income swings. Particularly useful when pension income provides a comfortable floor — the small nominal spending dip in loss years is easily absorbed. Can be combined with Guardrails for additional protection.

Modifier 2: Guardrails (Guyton-Klinger)

Guardrails dynamically adjust your spending up or down based on how far your current withdrawal rate has drifted from your initial withdrawal rate. The initial withdrawal rate is calculated at retirement as base spending divided by total portfolio value. Each year from year 2 onwards, the calculator compares your current withdrawal rate (current spending base divided by current real portfolio value) to this initial rate. If it has drifted too far, spending is adjusted.

How it works:

1. Upper guardrail trigger: If your current withdrawal rate falls more than the upper threshold (default 20%) below the initial rate — meaning the portfolio has grown faster than spending — spending increases by the adjustment amount (default 10%). This prevents the common trap of underspending while sitting on a large portfolio.
2. Lower guardrail trigger: If your current withdrawal rate rises more than the lower threshold (default 15%) above the initial rate — meaning the portfolio has fallen behind spending — spending is cut by the adjustment amount (default 10%). This protects portfolio longevity during downturns without requiring a dramatic lifestyle change all at once.
3. Pension floor protection: Spending can never fall below total pension income (defined benefit + Age Pension), ensuring a minimum income regardless of market performance.

Example: Starting with \$1.5M portfolio and \$90,000 spending = 6.0% initial withdrawal rate. Upper trigger at 20% below = 4.8%. Lower trigger at 15% above = 6.9%.

- Year 6: Strong markets, portfolio \$1.9M, spending \$96k = 5.1% WR. Still above the 4.8% upper trigger — no change.
- Year 8: More strong returns, portfolio \$2.2M, spending \$100k = 4.5% WR. This is now below the 4.8% upper trigger — spending increases by 10% to \$110k. The calculator recognises you can afford more.

- Year 12: Market downturn, portfolio drops to \$1.3M, spending \$115k = 8.8% WR. Above the 6.9% lower trigger — spending cut by 10% to \$103.5k. The portfolio is protected from further depletion.

Configuration Parameters:

- Upper Guardrail (default 20%): How far below the initial WR triggers a spending increase. Higher values (30%) trigger increases more readily — you get raises more easily but also risk overspending if the trigger fires too early.
- Lower Guardrail (default 15%): How far above the initial WR triggers a spending cut. Lower values (10%) cut spending more readily, providing more portfolio protection but also more income volatility.
- Adjustment Amount (default 10%): The size of each spending change when a guardrail fires. Smaller adjustments (5%) produce smoother income changes but require more frequent triggers to achieve the same effect.

Advantages: Provides both upside and downside adjustment — unlike Forgo Inflation, guardrails give you spending increases when the portfolio performs well. Morningstar's 2025 research found a 5.2% starting safe withdrawal rate with guardrails, compared to 3.9% for a fixed real strategy. Lifetime spending is also higher than the fixed real base case. Pension income provides a natural floor that prevents spending from becoming unliveable in severe downturns.

Disadvantages: More complex than Forgo Inflation — requires understanding the parameters and monitoring. Year-to-year spending can be volatile; the simulation may show significant swings in the spending chart. The spending cuts in downturns require genuine lifestyle discipline. Not suitable if you have large fixed expenses that cannot be reduced (e.g. committed loan payments, fixed-term leases). Tends to produce lower ending portfolio balances than fixed-real strategies because the upside adjustments spend down more capital.

When to use: When you want to maximise lifetime spending and are comfortable with year-to-year variability. Most effective when a substantial portion of your essential living costs are covered by pension income (defined benefit + Age Pension) — the guardrail cuts then only affect discretionary spending, not necessities. Well suited to retirees who are financially sophisticated and actively monitor their position. Consider starting with the default parameters (20/15/10) and adjusting once you have seen how the simulation behaves across different scenarios.

Combining Strategies

The two modifiers are independent and can be used together. When both are active, Forgo Inflation runs first (adjusting the real spending base downward in loss years), then Guardrails compares the resulting withdrawal rate to the initial rate and adjusts further if needed. The four possible combinations are:

- Neither modifier: Pure spending pattern (Constant Real or J.P. Morgan). Maximum predictability, lowest longevity protection. Good for baseline modelling.

- Forgo Inflation only: Small, automatic, cumulative protection. Low volatility. No configuration needed. Recommended as a default for most retirees.
- Guardrails only: More active management. Provides spending increases in good years and cuts in bad years. Higher expected lifetime spending but more year-to-year volatility.
- Both modifiers: Maximum longevity protection and maximum lifetime spending potential. In loss years, spending is first held flat (Forgo Inflation), and then if the withdrawal rate has also drifted above the lower guardrail, it is cut further. In strong years, the guardrail upper trigger can fire to increase spending. Most complex to track but most responsive to actual portfolio performance.

Australian Minimum Super Drawdown

Regardless of which strategy you choose, superannuation account-based pensions must pay a minimum percentage of the account balance each year under ATO regulations. These minimums are enforced in the simulation as a hard floor that operates independently of any spending decision:

- Under 65: 4% • Ages 65–74: 5% • Ages 75–79: 6% • Ages 80–84: 7% • Ages 85–89: 9% • Ages 90–94: 11% • Age 95+: 14%

In practice this means that in later retirement — particularly beyond age 80 — the minimum drawdown requirement will often exceed what your chosen withdrawal strategy calls for. When this occurs, the simulation releases the minimum required amount from super into a cash account, from which it is available to fund spending. If spending is lower than the minimum drawdown in a given year, the surplus accumulates in the cash account at the configured Cash/Buffer Return Rate (default 3%) and carries forward. The minimum drawdown is not a spending mandate — only a release requirement. Any unspent amount remains available

3.2 One Off & Irregular Expenses

The calculator offers two complementary approaches to modelling expenses outside your regular annual spending: Stochastic Irregular Expenses (recommended) and Manual One-Off Expenses. Both can be used simultaneously.

Stochastic Irregular Expenses (Recommended)

Stochastic irregular expenses model the unpredictable but inevitable costs of retirement — vehicle replacements, home maintenance, medical expenses, and accessibility modifications. Rather than guessing exactly when your roof will need replacing, the model uses probability-based timing and cost variation to generate realistic expense paths.

Included expense categories:

- Transport: Vehicle replacement approximately every 10 years (\$50k base, varies $\pm 15\%$)
- Housing: Roof, HVAC, painting cycles plus emergency repairs (1.5% annual probability)
- Medical: Dental work, hearing aids, hospital excess (increases with age)
- Home modifications: Accessibility updates (age-dependent probability, one-time \$30k base)
- Expected range: ~\$7-10k/year median, ~\$12-16k/year at the 90th percentile

Note: Aged care costs are not included in stochastic expenses. Use the dedicated Aged Care section for modelling RAD deposits, annual fees, and refunds.

How it works across test scenarios:

- **Constant Return / Historical Period:** Generates one deterministic expense path using a fixed seed. Results are reproducible — you'll see the same expenses in the same years every time. This provides a realistic "middle-ground" scenario for quick planning.
- **Monte Carlo:** Generates a different expense path for each of the 1,000 simulations. Each run has unique timing and amounts, capturing uncertainty in both market returns and irregular expenses. This is the recommended approach for comprehensive planning.
- **Historical Monte Carlo:** Generates a different expense path for each simulation combined with actual historical return sequences, providing maximum realism.

Reshuffling expense timing: The  Reshuffle Expense Timing button generates a new randomised expense path by changing the seed. Each seed produces a different but reproducible pattern — useful for checking whether your plan is sensitive to expense timing without running a full Monte Carlo. The seed number is displayed and can be reset to the default at any time.

Manual One-Off Expenses

Manual one-off expenses are for specific planned expenditures at known ages — a gift to family at 65, a major holiday at 70, paying off the mortgage at 62. These are expenses you can reasonably predict the timing and cost of.

Important: Amounts are entered in nominal (future) dollars — the dollar value in the year the expense occurs, not in today's dollars. For example, \$50,000 for a car at age 70 means \$50,000 in age-70 dollars. If you are estimating in today's dollars, add roughly 2.5% per year for inflation to convert. This differs from base spending, which is entered in retirement-year dollars and automatically inflated each year.

No default expenses are pre-loaded. Add items as needed using the "+ Add One-Off Expense" button, specifying a description, the age it occurs, and the amount.

Using Both Together

Stochastic and manual expenses can run simultaneously. When both are enabled, a warning reminds you to avoid double-counting: stochastic expenses already cover vehicle replacements, home maintenance, and medical costs, so manual entries should be limited to planned expenses outside those categories (e.g., gifts, holidays, mortgage payoff, lifestyle purchases).

With Monte Carlo simulations, note that manual one-off expenses use the same fixed amounts in every run, while stochastic expenses vary across simulations.

3.3 Aged Care Modelling

Cost Components

RAD (Refundable Accommodation Deposit): Lump sum paid on entry (typical: \$400k-\$550k). Refunded when exiting care. Withdrawn from, and refunded to, super balance.

Annual Care Costs: Daily fees, means-tested fees, extra services (typical: \$60k-\$80k/year). Indexed to inflation.

Modeling Approaches

Deterministic: Enter specific age and duration. Use for planned entry scenarios.

Probabilistic: Risk-based entry using actuarial data. Use for Monte Carlo simulations to model uncertainty.

Couple Mode Integration

In couple tracking, aged care is modelled for the surviving partner only; the calculator does not currently support modelling one partner entering care while both are alive. Both partners are assumed healthy until a death event occurs. After first partner dies, the survivor may enter care at specified age.

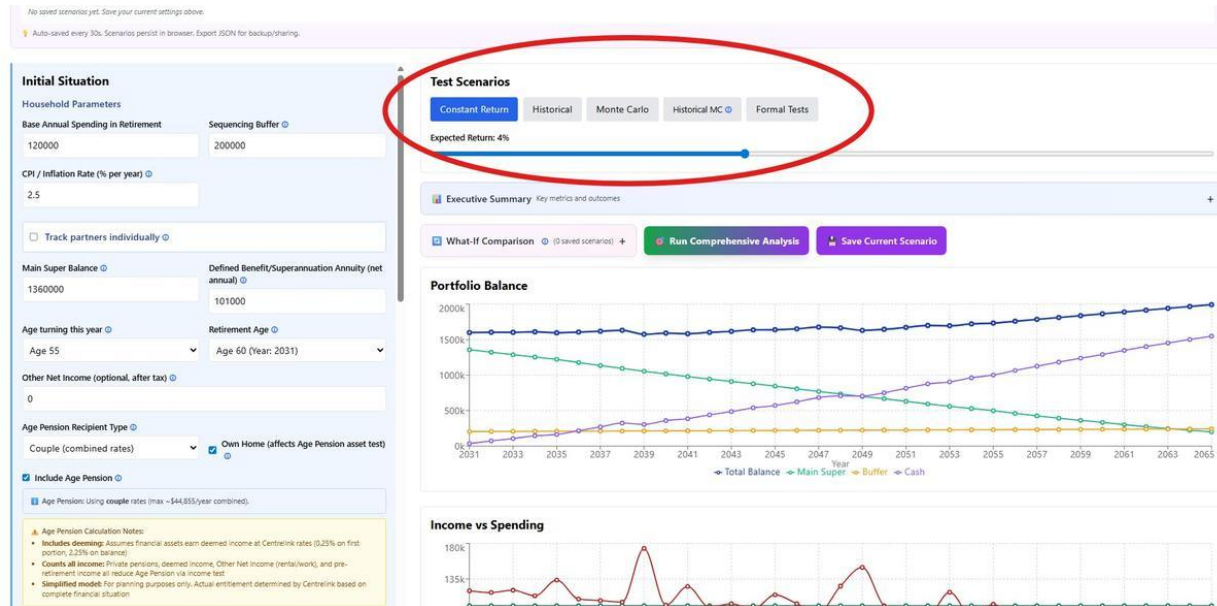
3.4 Debts & Liabilities

Models mortgages, personal loans, or other debts carried into retirement. For each debt, enter the outstanding balance, interest rate, and repayment term. Interest is calculated monthly and payments are applied to principal first. The balance cannot go negative — any overpayment in the final period is automatically capped.

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PART 4: RUNNING SCENARIOS

4.1 Test Scenario Types



A Note on Fees

The return rate used in the Constant Return, Monte Carlo, and Historical Monte Carlo Test Scenarios is net of fund or investment fees. To account for investment fees, deduct them from the gross return (e.g. gross 7.5% minus 0.5% fees = 7% Expected Return).

Constant Return

Fixed annual return (e.g., 5%). Simplest scenario for baseline testing. Useful for: initial planning, comparing to rules of thumb, understanding basic dynamics. **Note:** The Constant Return slider represents **NOMINAL** returns (before adjusting for inflation).

The simulation always runs in **NOMINAL** dollars with your selected CPI rate built in.

How it works

If you select a 5% constant return with 2.5% CPI:

- Main Super grows at 5% **NOMINAL** each year
- But spending increases by 2.5% (inflation)
- Net **REAL return** = 5% - 2.5% = 2.5% real

Monte Carlo Simulation

1,000+ scenarios using lognormally distributed investment returns, generated from normally distributed random draws using the Box–Muller transform. Four asset allocation presets provide starting-point assumptions: Conservative (~30% equities, 6.5% return, 8% vol), Balanced (60/40, 7.5% return, 11% vol), Growth (70/30, 8.0% return, 13.5% vol), and High Growth (~90% equities, 9.0% return, 18% vol). You can also set a Custom return and volatility. The Balanced preset (7.5%/11%) is the default. Provides success rate and outcome range (P10–P90). Optional: enable “Variable death age in MC” to sample a different death age per run from ABS life tables (see Section 2.1), producing a more realistic spread of outcomes that reflects genuine longevity uncertainty. Use for: uncertainty quantification, risk assessment, confidence levels.

Historical Monte Carlo

Randomly samples verified historical return series rather than assumed distributions. Four data series are available:

- AU Balanced (60/40, 1990–2024, 35 years): 60% XAOA / 40% AusBond Composite. Approximates a typical Australian industry super balanced option. Includes GFC, 2022 bond crash, and COVID shock.
- AU Equities — XAOA (1980–2024, 45 years): ASX All Ordinaries Accumulation Index. Includes 1987 crash (-50%) and GFC (-40%).
- AU Bonds — AusBond Composite (1990–2024, 35 years): Australian investment-grade bonds. Lower volatility; captures the 1994 and 2022 rate shocks.
- US S&P 500 Total Return (1928–2025, 98 years): Shiller/Ibbotson dataset. The longest series; covers Great Depression, WWII, 1970s stagflation, Black Monday, and GFC.

Three sampling methods: Shuffled Years (random year sampling, destroys autocorrelation), Block Bootstrap (overlapping blocks, default 5-year blocks — recommended), and 35-Year Blocks (contiguous 35-year periods, S&P 500 only). Combines historical accuracy with Monte Carlo uncertainty. Generally more conservative than parametric Monte Carlo for AU-series data.

Longevity & Outcome Analysis (Scatter Chart)

When Parametric or Historical Monte Carlo is active and results have been run, a Longevity & Outcome Analysis chart appears immediately below the Executive Summary. It provides two views via tabs:

Longevity scatter: Each dot is one MC run. The X axis shows the simulation end age; the Y axis shows the final portfolio balance. Green dots are solvent runs; red dots are insolvent. The scatter reveals whether failures cluster at long lifespans (longevity risk) or at average ages regardless of how long the person lived (return-driven risk). When stochastic mortality is on you will see genuine spread on the X axis; when it is off all dots share the same end age.

Ruin age distribution: Shows only failed runs, binned by the age at which the portfolio reached zero. This is more actionable than the overall success rate: it tells you whether failure typically

occurs at age 82 or age 98, which determines whether a modest spending reduction or a fundamental plan change is needed.

The chart hides automatically when any non-MC scenario type is selected (Constant Return, Historical Period, or Formal Tests).

Stress Tests

Structured scenarios testing specific failure modes. Unlike Monte Carlo, which tests probability across thousands of random paths, formal stress tests probe specific vulnerabilities in a controlled way — each test isolates one risk factor so you can see exactly how your plan responds to that scenario.

Baseline

- **A1 Base Case:** 5% constant nominal return throughout. The reference scenario against which all other tests are compared. If your plan fails A1, it has a fundamental structural problem.
- **A2 Low Returns:** 3.5% constant nominal return. Models a structural low-growth environment — persistent low rates, an overvalued starting market, or a conservative portfolio. A plan that passes A1 but fails A2 is sensitive to return assumptions and should be stress-tested further with Monte Carlo.

Market Shock

- **B1 Crash:** -25% in year 1, -15% in year 2, then 5% constant. Tests the sequence-of-returns risk from an immediate severe market crash at the point of retirement — the worst timing possible, when the portfolio is at its largest.
- **B2 Bear Market:** 0% returns for 10 years, then 5% constant. Tests extended stagnation rather than a sharp crash. Particularly relevant for bond-heavy portfolios and those relying heavily on the sequencing buffer.
- **B3 High Volatility:** Alternating positive and negative returns averaging 5% nominal. Tests volatility drag — the mathematical reality that a portfolio experiencing +20% then -20% does not return to its starting value, even though the arithmetic average is 0%.

Inflation

- **C1 High Inflation:** 5% CPI throughout retirement. Tests purchasing power erosion over a long horizon. At 5% CPI, your spending doubles in real dollar terms approximately every 14 years. Plans that rely heavily on fixed-income sources (non-indexed pensions, fixed annuities) are most exposed.

Longevity

- **D1 Extreme Longevity:** Simulation runs for 45 years to age 105, overriding the Modelled Death Age. Tests whether your plan can sustain a significantly longer-than-expected

retirement. A plan that fails D1 but passes at a shorter horizon has longevity risk that Monte Carlo with variable death age will also reveal.

Health

- **G1 Health Shock:** Additional \$30,000 per year in health-related costs from age 75, on top of regular spending and any aged care already modelled. Tests the impact of chronic health costs in later retirement without triggering the full aged care model.

Combined

- **H1 Worst Case:** Combines the B1 crash, C1 high inflation, and G1 health shock simultaneously. This is a low-probability but not impossible scenario — a market crash at retirement followed by a decade of high inflation and elevated health costs. A plan that survives H1 is genuinely robust.

A robust plan should pass A1, A2, and at least B2. Passing B1 and H1 indicates a high degree of resilience. Use the results to identify which specific risk your plan is most exposed to, then address that risk directly — for example by increasing the sequencing buffer if B1 is the failure point, or by ensuring sufficient indexed income if C1 is the weak spot.

Historical Periods

Tests your retirement plan against specific historical return sequences. A two-step selector lets you choose the data source first, then pick the starting period. All AU periods use genuine historical data throughout. US S&P 500 periods are divided into two categories: three are genuine 35-year historical replays using real data throughout, and three are stress tests that open with a real crash but cannot complete a full 35-year window because the event occurred too recently — labelled [Stress Test] in the selector.

Data Sources

- US S&P 500 (1928–2025): Shiller/Ibbotson total return dataset. The longest available series; covers the Great Depression, WWII, 1970s stagflation, Black Monday, and the GFC.
- AU Equities — XAOA (1980–2024): ASX All Ordinaries Accumulation Index. Covers the 1987 crash (–50%), 1990 recession, and GFC (–40%).
- AU Balanced 60/40 (1990–2024): 60% XAOA / 40% AusBond Composite proxy. More realistic for typical Australian diversified super funds.
- AU Bonds — AusBond Composite (1990–2024): Australian investment-grade bonds. Reference scenario only — not a stress test.

Available Periods (11 total)

US S&P 500 — Genuine 35-year replays (3 periods): 1929 Great Depression (three consecutive losses: –8%, –25%, –43%); 1973 Stagflation (–15% then –26% with decade-long high inflation);

1982 Bull Market (strongest sustained bull market; best-case upper bound). All use 35 years of real S&P 500 data.

US S&P 500 — Stress Tests with real opening crash (3 periods): 2008 GFC [Stress Test] (17 real years 2008–2024, then flat 10%/yr); 2000 Dot-com [Stress Test] (25 real years 2000–2024, then flat 10%/yr); 2020 COVID [Stress Test] (5 real years 2020–2024, then flat 10%/yr). These periods cannot complete 35 years of real data — a 2008 start needs data to 2043, which does not yet exist. The tail years use flat 10%/yr nominal (the long-run S&P 500 average) as a neutral assumption. Note: the 2020 opening figure of –18% reflects an intra-year trough — the 2020 full-year S&P 500 return was actually +18%.

AU Equities (3 periods): AU 1990 Recession — hardest AU stress test, opens –17.5% with GFC in year 19; AU 1987 Crash — opens –7.9% with 1990 recession in year 4; AU 1980 Bull Market — best-case comparator, not a stress test.

AU Balanced (1 period): AU Balanced 1990 Recession — opens –10.1%, covers GFC (–18.2%) and 2022 rate shock.

AU Bonds (1 period): AU Bonds 1990–2024 — reference only; bonds rallied as rates fell over this period.

Periods are labelled by the financial event that opens the sequence rather than by neutral year, making it clear which scenarios are genuine stress tests and which are best-case comparators. Each period shows a contextual description beneath the selector explaining which events the sequence captures and, for [Stress Test] periods, how many years of real data are used. The [Stress Test] periods remain useful for testing sequence-of-returns risk: the opening crash is real, and the flat 10% tail represents average market conditions thereafter rather than an optimistic recovery.

4.2 Understanding Results

Success Criteria

A simulation is successful if the portfolio balance remains at or above zero at the Modelled Death Age (or the formal test target age for stress tests). A plan that ends with exactly \$0 counts as a success — perfect timing. A plan that depletes one year before the target age is a failure, regardless of how close the ending balance was to zero.

Interpreting Monte Carlo Success Rates

The success rate is the percentage of simulations in which the portfolio lasted the full retirement period. It is not a guarantee — it is a probabilistic estimate based on your inputs and assumptions.

As a rough guide:

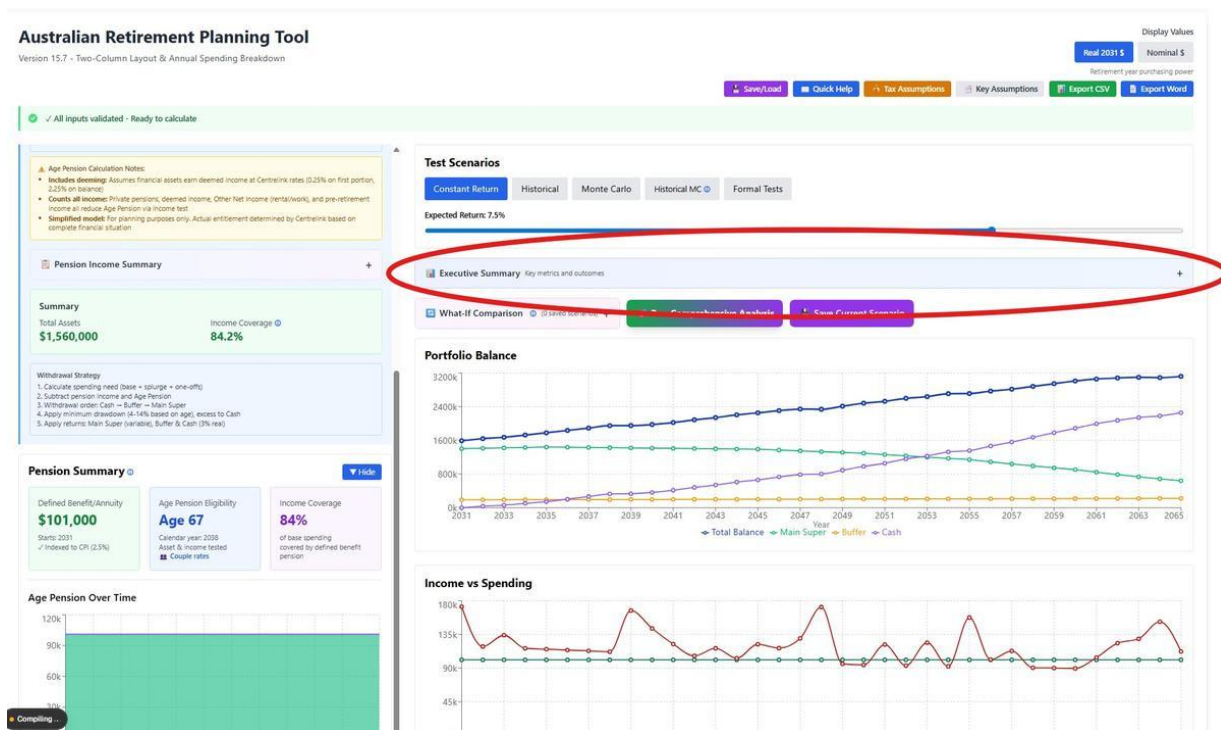
Success Rate	Parametric MC	Historical MC
90%+	Excellent — very robust plan	Excellent — survived nearly all of history
80–89%	Good — meets industry standard	Good — survived most periods including major crashes
70–79%	Moderate — consider adjustments	Moderate — some challenging periods caused failure
Below 70%	Needs attention	Needs attention

Historical MC has higher thresholds because it is bounded by real events — the worst year in the dataset is –43% (1931). Parametric MC draws from a statistical distribution that can generate scenarios worse than anything in the historical record, producing lower success rates for the same portfolio. This means Historical MC typically shows 5–15 percentage points higher success rates than Parametric MC. Both numbers are valid — they answer slightly different questions.

Use Parametric MC when you want to stress-test against theoretical tail risks beyond history. Use Historical MC when you want to know how your plan would have fared against the actual market record. Ideally run both and treat the gap between them as an indicator of your plan's sensitivity to extreme scenarios..

4.3 Executive Summary Dashboard

A collapsible panel above the charts provides comprehensive at-a-glance analysis. Click '  Executive Summary' to expand.



Components

The Executive Summary is organised into several panels, each collapsing to a header when not needed.

Scenario Indicator — shows which test scenario is currently active and, for Monte Carlo modes, a reminder that all figures shown reflect the median (P50) simulation path rather than the best or worst case.

Success/Fail Banner — a green  or red  banner showing whether the portfolio lasted the full modelled horizon. Includes the number of years lasted, the ending balance, and context-aware messaging that varies based on the scenario type and outcome. In Monte Carlo mode the banner also shows the overall success rate across all simulations.

Portfolio Summary — starting balance, ending balance, total amount withdrawn from the portfolio, average annual return, and safety margin (ending balance as a percentage of starting balance).

Income Summary — private pension income at retirement, the year Age Pension payments first appear, maximum Age Pension received in any single year, average annual spending, and peak spending year.

Risk Assessment — four traffic light indicators covering sequence-of-returns risk (were there negative returns in the critical early years?), longevity risk (did the portfolio run low in later retirement?), inflation risk (is the CPI rate elevated?), and health costs (are aged care costs modelled?).

Monte Carlo Analysis — success rate and P10/P50/P90 final balance outcomes. This panel only appears when Monte Carlo or Historical Monte Carlo is active.

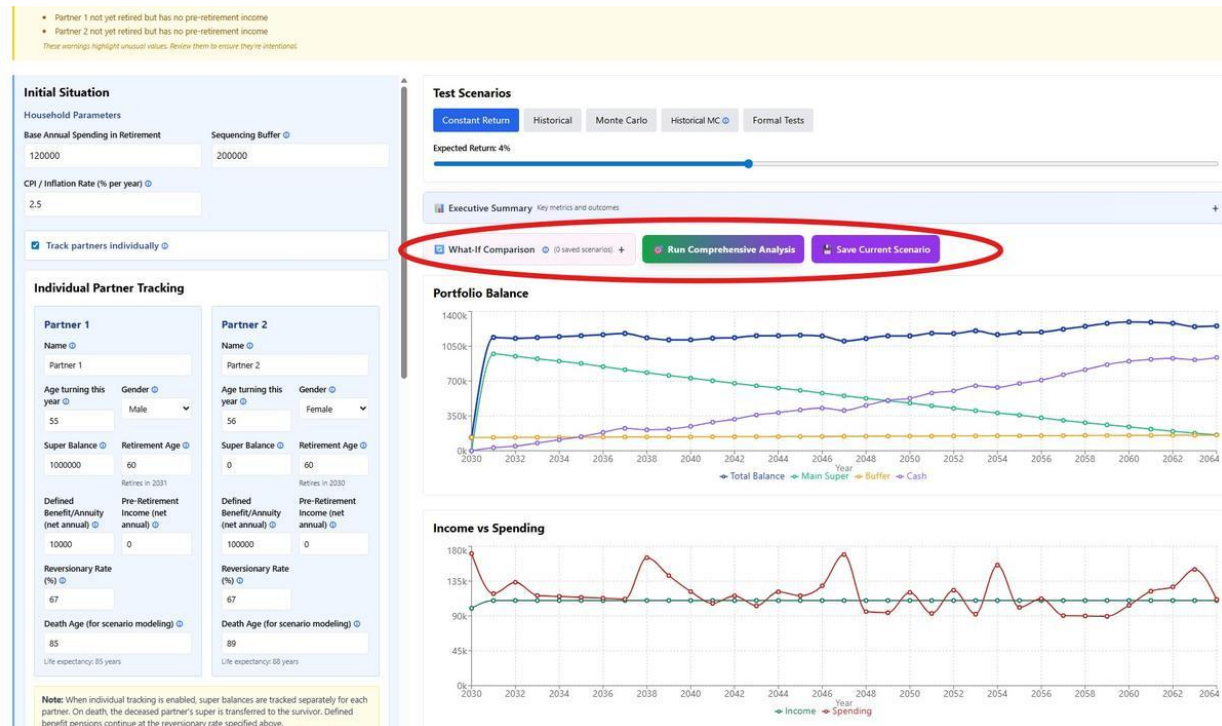
Individual Partner Summary — final superannuation balance and pension income for each partner. Only appears in couple tracking mode.

Every metric has a hover tooltip (blue ⓘ icon) explaining what it measures and how to interpret it. All values respect the Real/Nominal display toggle at the top of the screen.

PART 5: WHAT-IF SCENARIO COMPARISON

5.1 Overview

The What-If Scenario Comparison feature allows you to save and compare up to 5 different retirement scenarios side-by-side. This helps you understand how changes in key parameters (super balance, spending, retirement age, returns) affect your retirement outcomes.



5.2 Basic Workflow

To use What-If Comparison:

1. Set up your baseline scenario with initial parameters. Run the applicable Test Scenario.
2. Click "Save Current Scenario" to capture the baseline
3. Modify one or more parameters (e.g., reduce spending by \$10k/year). Rerun the Test Scenario.
4. Click "Save Current Scenario" again to save the variant
5. Repeat for up to 5 total scenarios
6. Compare results side-by-side in the comparison table

The comparison table shows both key results (Success/Fail, Ending Balance, Years Lasted, MC Success Rate, Formal Tests Passed) and input parameters for each scenario.

5.3 Comprehensive Analysis

The "Run Comprehensive Analysis" button (located at the top of the What-If panel) provides complete risk assessment by running both Parametric Monte Carlo AND all 9 Formal Stress Tests in a single operation.

What it does:

- Runs 1,000 parametric Monte Carlo simulations using your Expected Return & Volatility settings
- Runs all 9 Formal Stress Tests (crash scenarios, longevity tests, inflation tests, etc.)
- Automatically saves a scenario with both MC success rate and formal tests results
- Takes approximately 5-10 seconds to complete

This is ideal when you need complete risk assessment across both statistical probability (Monte Carlo) and specific failure modes (Formal Tests). For faster comparisons using only one test type, use the individual scenario buttons (Constant Return, Historical, Monte Carlo, or Formal Tests) with the "Save Current Scenario" button.

5.4 Understanding the Comparison Metrics

The comparison table displays five key result metrics:

- **Success/Fail:** Whether the portfolio lasted to target age with balance $\geq$ \$0
- **Ending Balance:** Portfolio value at end of simulation (in real or nominal dollars as per display setting)
- **Years Lasted:** How many years the portfolio sustained spending
- **Parametric MC Success Rate:** Percentage of parametric Monte Carlo simulations that succeeded (uses Expected Return $\pm$ Volatility). Shows "—" if not applicable
- **Formal Tests Passed:** Number of formal stress tests passed out of total (X / Y format). Shows "—" if not applicable

Green ▲ indicators show improvement over the current scenario, red ▼ shows worse performance, and gray = shows equal performance.

5.5 Key Parameters to Vary

Common parameters to test in What-If scenarios:

- **Super Balance:** Test the impact of higher or lower starting superannuation balances
- **Base Spending:** Compare more frugal versus more generous lifestyle spending
- **Retirement Age:** Compare retiring at different ages (e.g., 60 vs 65)
- **Defined Benefit/Annuity:** Model different pension income amounts
- **Splurge Spending:** Test major one-time expense scenarios
- **Return Scenarios:** Compare optimistic vs pessimistic return assumptions (Constant, Historical, Monte Carlo)

5.6 Managing Saved Scenarios

Scenario management features:

- **Rename:** Click on any scenario name to edit it (e.g., "Conservative", "Aggressive", "Base Case")
- **Delete:** Click the  button next to individual scenarios to remove them

- **Clear All:** Remove all saved scenarios at once with the "Clear All" button
- **Maximum:** Up to 5 scenarios can be saved simultaneously - delete one to add more

The help section within the What-If panel can be collapsed/expanded using the + / – button for a cleaner interface.

5.7 Sensitivity Analysis — Return vs Spending

Overview

The Sensitivity Analysis table shows how your ending portfolio balance at your Modelled Death Age changes across a range of investment return assumptions and spending levels. It provides an at-a-glance map of your plan's resilience — how much margin you have before the plan fails, and which direction (lower returns vs higher spending) poses the greater risk.

Availability: The Sensitivity Analysis panel is only shown when Constant Return is the active scenario. It is hidden in Historical Period, Monte Carlo, Historical Monte Carlo, and Formal Test modes — those modes either model specific return sequences or uncertainty directly, making a deterministic return grid redundant. Switch to Constant Return to access it.

The table is located in the interface between the Age Pension Details panel and the What-If Comparison panel. Click the “ Sensitivity Analysis — Return vs Spending” header to expand or collapse it.

Reading the Table

The table is a grid with **investment return on the vertical axis** (rows) and **spending level on the horizontal axis** (columns). Return rows span your current setting $\pm 3\%$ in 1% steps (capped at 2%–15%). Spending columns show 80%, 90%, 100%, 110%, and 120% of your current base spending, with the dollar amount shown in each column header.

Each cell shows the ending portfolio balance (or depletion year if the plan fails). Your current settings are highlighted with a blue border. The colour coding indicates the outcome:

Bright green: Survived to Modelled Death Age with more than \$1M remaining (in the active Real or Nominal view)

Light green: Survived with \$500k–\$1M remaining

Yellow: Survived but less than \$500k remaining (thin margin)

Orange: Portfolio depleted after year 25 (late-stage failure)

Red: Portfolio depleted before year 25 (serious shortfall)

The colour thresholds adjust automatically when you switch between Real and Nominal dollar views, so the green/amber/yellow boundaries stay meaningful in both views.

What the Table Includes and Excludes

Each cell varies only the investment return and base spending level. Everything else is drawn from your current settings:

Included (fixed at your current settings): All income streams (other income, defined benefit pension, Age Pension), aged care, couple tracking, partner death scenarios, debt repayment, splurge spending, guardrails, and inflation rate.

Excluded: Stochastic irregular expenses are not modelled (the random expense path is omitted for deterministic consistency). Market volatility is also excluded — each row uses a constant return for the full simulation horizon. For volatility risk, use Monte Carlo.

Why the Blue Cell May Differ from the Executive Summary

The blue cell (your current return and 100% spending) shows the single deterministic constant-return outcome — the same result you would see if you selected Constant Return mode in the calculator. The Executive Summary may show different values if Monte Carlo or Historical mode is active, because those modes show median results across many simulations rather than a single deterministic run. Stochastic irregular expenses, if enabled, will also create a small difference since the sensitivity table excludes them.

The sensitivity table is best understood as a “what if returns were exactly X% every year” grid. It answers the structural question of whether your plan is viable at different return and spending assumptions, while Monte Carlo answers the probabilistic question of how likely different outcomes are given market volatility.

Word Export

The sensitivity grid is included in every Word export as a full-colour table on its own page, appearing between the Detailed Results section and the Assumptions appendix. The colour coding (green, amber, yellow, orange, red) is reproduced in the Word document. The grid always reflects the settings in effect at the time of export.

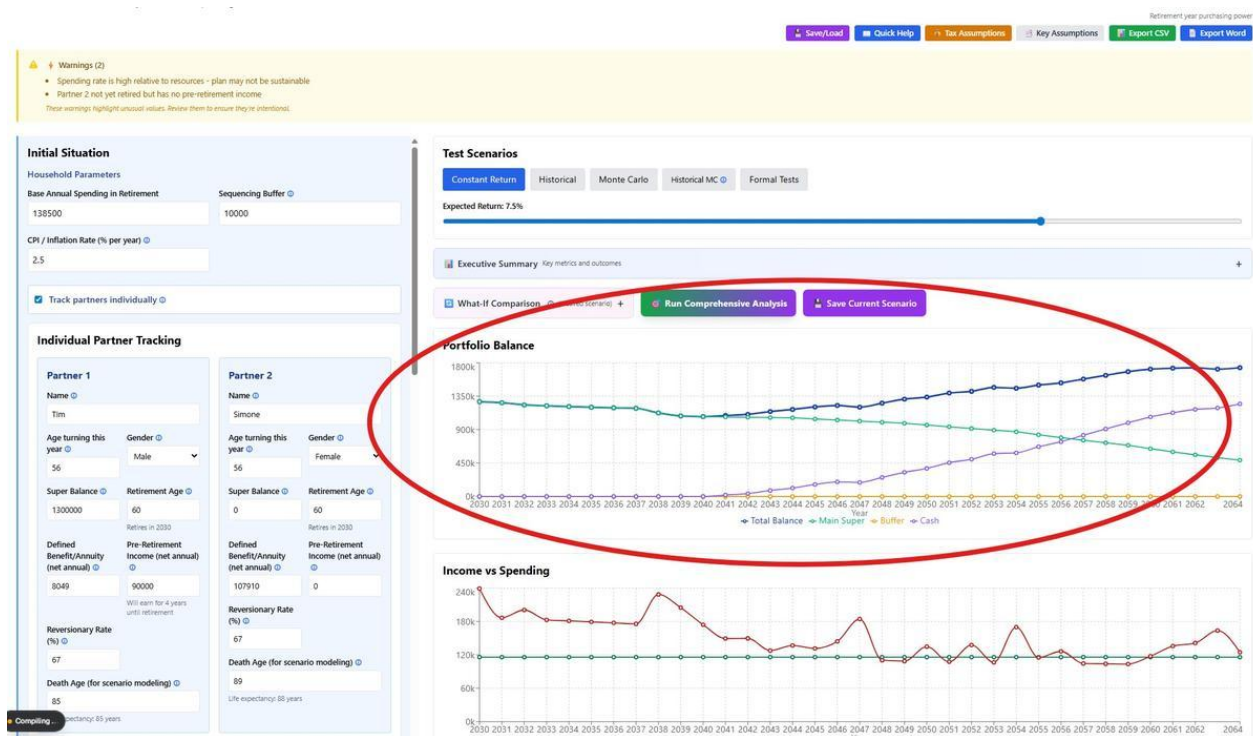
PART 6: CHARTS & ANALYSIS

6.1 Chart Types & Interpretation

Portfolio Balance Chart

Shows total portfolio value over time. In standard scenarios (Constant Return, Historical Period, Formal Tests), four lines are displayed: Total Balance (blue), Main Super (green), Sequencing Buffer (orange), and Cash Account (purple). The component lines (Super, Buffer, Cash) hide automatically when percentile bands are toggled on, to avoid visual clutter.

In Monte Carlo mode, only the Total Balance median (P50) line is shown alongside the percentile bands. Toggle the "Show/Hide Bands" button to display or hide the P10/P25/P75/P90 boundary lines. When Variable death age is enabled, the bands extend beyond the median run's end age — reflecting the full distribution of outcomes across runs that ended at different ages — while the median line stops at the median run's death age. The income and spending bands in the Income vs Spending chart narrow progressively at older ages as fewer runs remain active.



Income vs Spending Chart

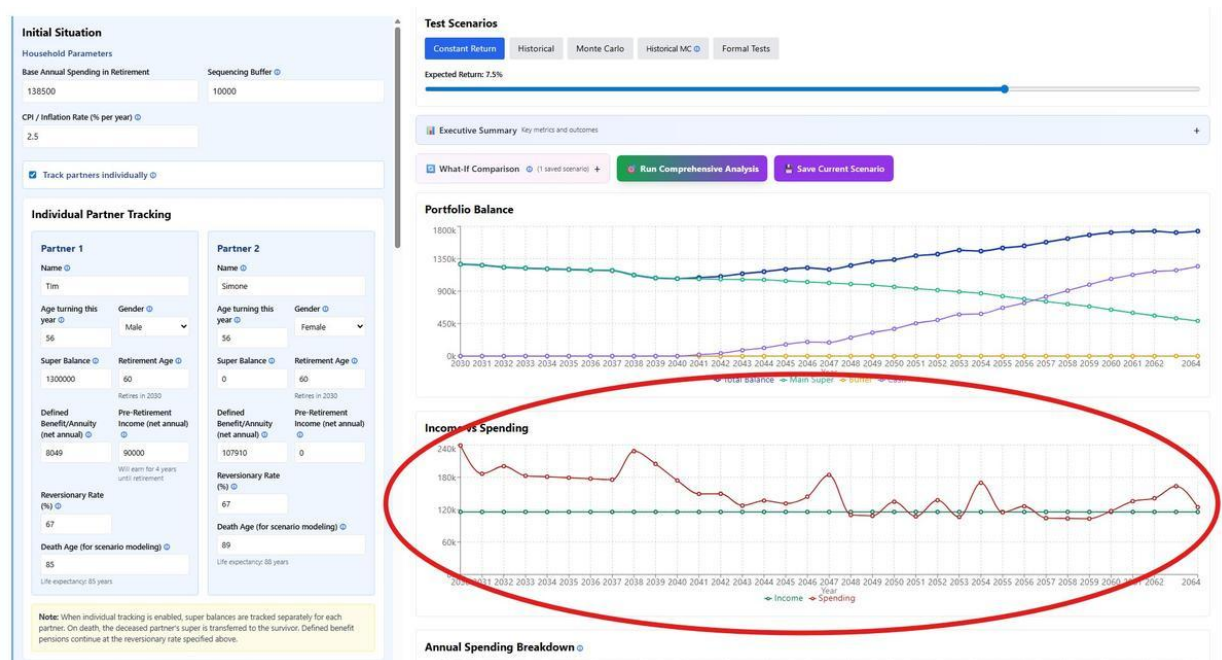
Shows annual income and spending across the retirement horizon on the same chart, making it easy to see how the gap between them evolves over time.

Income sources shown in green include defined benefit or annuity pension, Age Pension, other net income, and any additional income streams. Spending shown in red includes base spending (adjusted for the selected spending pattern), splurge spending, aged care costs, and debt

repayments. The gap between income and spending in any given year represents the amount drawn from the portfolio — a widening gap indicates increasing reliance on capital drawdown.

In Monte Carlo mode, the chart shows the median (P50) income and spending paths alongside percentile bands. The bands reflect only runs that are still active at each age — see the Portfolio Balance chart description for details on how the bands behave when Variable death age is enabled.

A healthy plan shows income covering an increasing share of spending over time as the Age Pension phases in and portfolio withdrawals fall. A plan where the income line stays flat or declines while spending rises is drawing down capital at an accelerating rate and warrants closer attention.



Annual Spending Breakdown Chart

Shows the composition of annual spending by funding source as a stacked bar chart, revealing not just how much you are spending each year but exactly where the money is coming from.

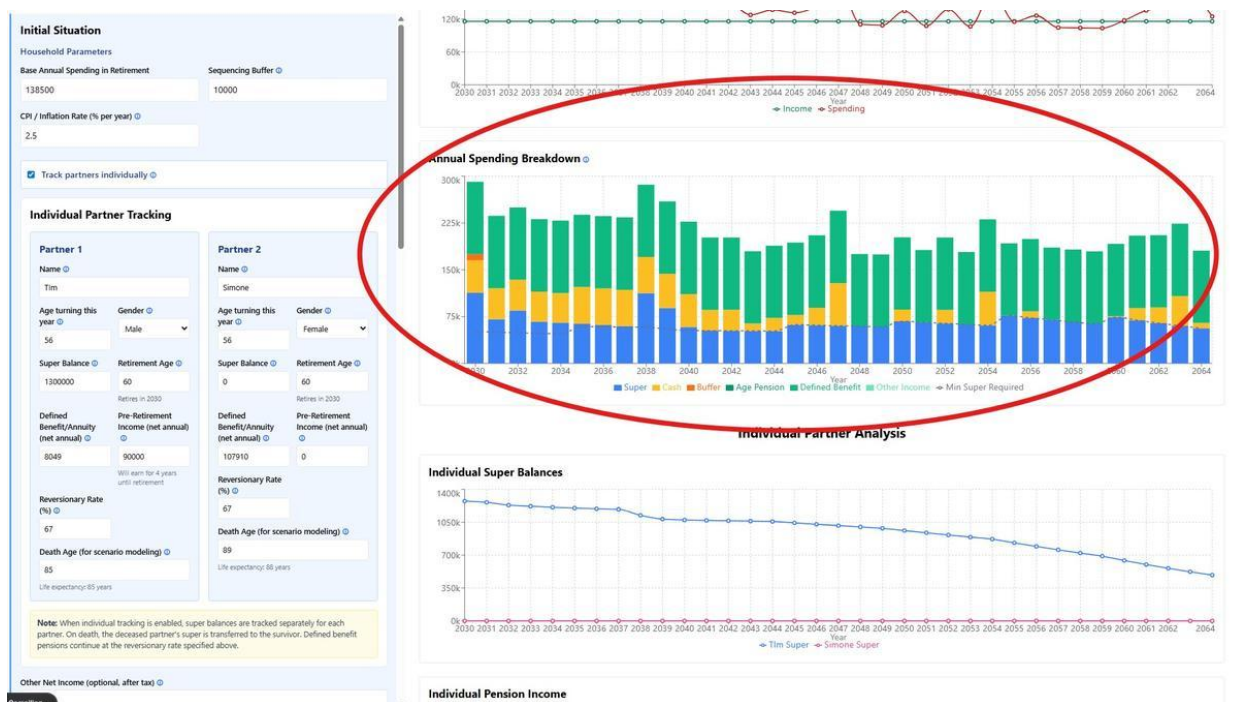
Bars are stacked from bottom to top in withdrawal priority order: Super withdrawals (blue), Cash Account (yellow), Sequencing Buffer (orange), Age Pension (dark green), Defined Benefit or annuity pension (medium green), and Other Income (light green). A bar dominated by blue Super at the top indicates heavy reliance on capital drawdown; a bar where the green income sources fill most of the height indicates a well-supported plan where pension income covers the majority of spending.

In Constant Return, Historical Period, and Formal Test modes, a gray dashed line overlays the chart showing the minimum required superannuation withdrawal for your age. The blue Super section

should meet or exceed this line each year to comply with ATO minimum drawdown requirements. This line is hidden in Monte Carlo modes.

In Monte Carlo mode, the chart shows the median (P50) simulation path only and ends at the median run's end age. A note below the chart explains this. For the full distribution of outcomes across all runs, refer to the Portfolio Balance chart above.

Key things to look for: the point at which Age Pension (dark green) first appears, typically around age 67; any sudden spike in bar height from a RAD payment on entering aged care; and whether the Cash and Buffer sections (yellow and orange) thin out and disappear over time as those accounts are drawn down.



This chart complements the Income vs Spending chart by showing not just the gap between income and spending, but exactly where the money comes from to fund your lifestyle.

Individual Super Balances (Couple Tracking)

Partner 1 (blue) and Partner 2 (pink) super balances tracked separately. Shows proportional withdrawals, death transfers, and depletion timeline.

Individual Pension Income (Couple Tracking)

Total pension for each partner: private pension + age pension allocation. Accounts for reversionary reductions and age pension rule changes.



Age Pension Calculation Details

A collapsible panel (labelled “🏠 Age Pension Calculation Details”) appears below the main charts when Age Pension is enabled and a simulation has been run. It provides a year-by-year table showing exactly how the Age Pension is calculated at each step, giving full transparency into the means-testing logic so you can verify the calculator’s output against Centrelink estimates or your own calculations.

Table Columns

The table displays one row per simulation year (from age 67, when Age Pension eligibility begins) with the following columns:

Year and Age: The simulation year number and corresponding age.

Rate: Whether couple or single Age Pension rates apply. This switches automatically from couple to single if a partner dies in couple tracking mode.

Assessable Assets: Total financial assets counted by Centrelink for the means test (Main Super + Sequencing Buffer + Cash Account + Assessable Assets Outside Super). All values are CPI-indexed. Your principal home is excluded if you are a homeowner.

Asset Threshold: The lower asset test threshold (CPI-indexed from 2025 base values). Assets below this threshold qualify for the full pension. Assets above it reduce the pension by \$3 per fortnight for every \$1,000 in excess.

Asset Test Pension: The pension amount resulting from the assets test. If assets exceed the cutoff, the pension is nil.

Deemed Income: Income that Centrelink assumes your financial assets earn, regardless of actual returns. Calculated at the lower deeming rate (0.25%) on the first portion of assets and the upper rate (2.25%) on the balance. Deeming thresholds are CPI-indexed.

Private Pension + Other: Combined non-deemed income sources counted in the income test, including: defined benefit/annuity pension, Other Net Income (rental, dividends, part-time work), and pre-retirement income if one partner is still working.

Total Testable Income: The sum of deemed income plus all non-deemed income. This total is compared against the income free area to determine the income test pension.

Income Free Area: Annual income you can earn before your pension starts to reduce. Single: \$5,512/yr; Couple: \$8,736/yr (combined). CPI-indexed each year. Income above this reduces the pension by 50 cents per dollar.

Income Test Pension: The pension amount resulting from the income test.

Binding Test: Centrelink applies both tests and pays the lower result. This column shows which test (Assets or Income) is binding. Highlighted in amber to draw attention to the constraint driving your pension amount. Shows “NIL” in red if assets exceed the cutoff threshold entirely.

Age Pension (Annual): The final pension amount paid per year. Shows “FULL” in green when receiving the maximum rate. Red rows indicate nil pension.

Reading the Table

All values respect the Real/Nominal dollars toggle, so you can view calculations in today's purchasing power or future nominal dollars. The most important column to watch is the Binding Test: if it consistently shows “Assets”, your portfolio balance is the main factor limiting your pension; if it shows “Income”, your defined benefit pension and other income streams are the constraint. As your portfolio draws down over time, you will typically see the binding test shift from Assets to Income, and the pension amount increase accordingly.

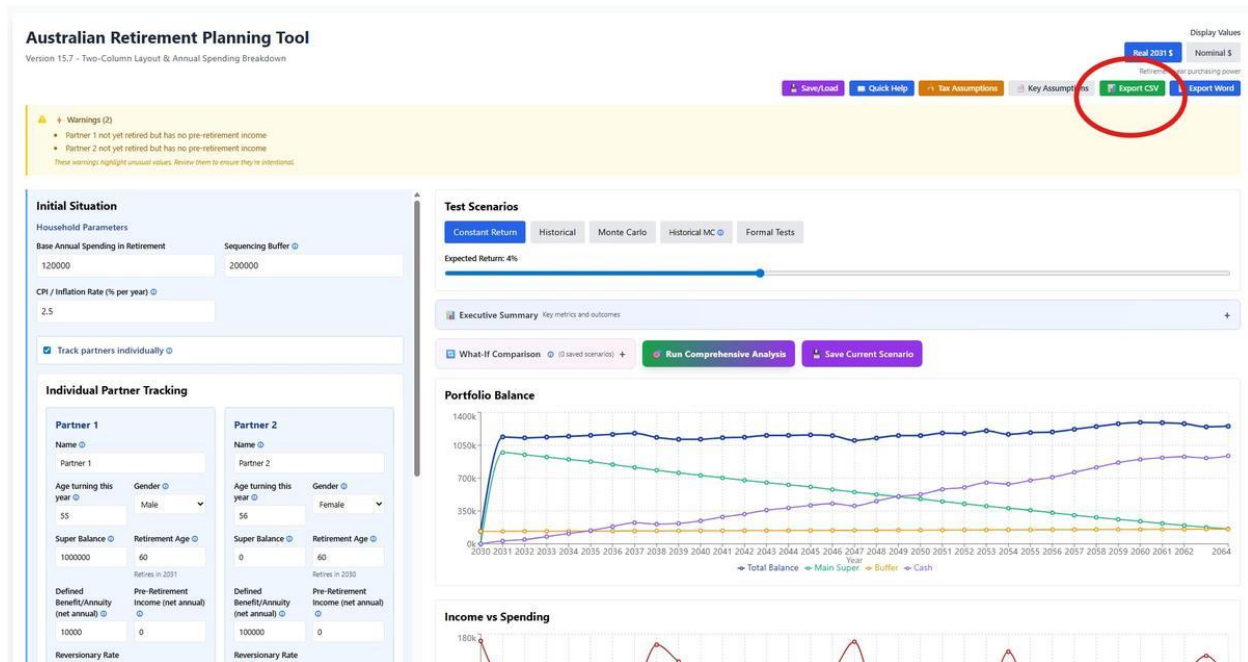
In Monte Carlo or Historical Monte Carlo mode, the table shows calculations for the median simulation path only. An amber warning reminds you that each of the individual simulations has different portfolio balances and therefore different Age Pension amounts. Treat the displayed values as a representative example, not a guarantee.

Practical Use

This panel is particularly useful for validating the calculator against your own Centrelink estimates, understanding when your Age Pension is likely to increase as assets draw down, identifying the crossover point where the binding test shifts from assets to income, and seeing the impact of non-super assessable assets on your pension entitlement. For a detailed explanation of the underlying calculation formulas, see Section 7.3 (Age Pension Calculation).

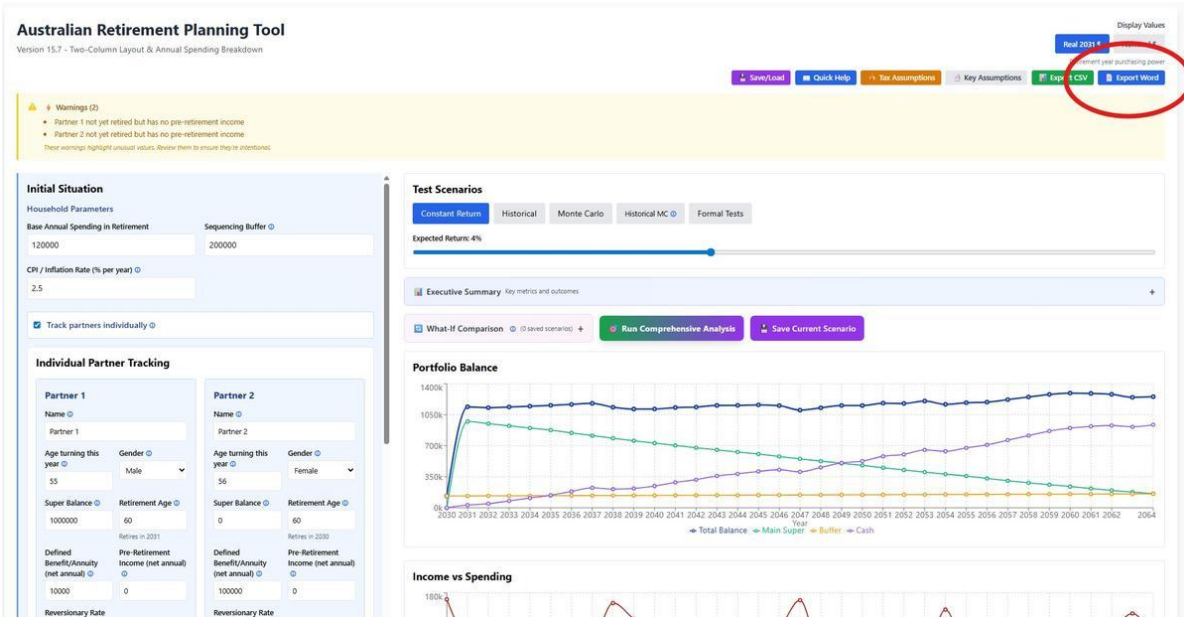
6.2 CSV Export & Analysis

Download detailed year-by-year data for external analysis. Includes: balances, income, spending, pensions, withdrawals, returns, ages, debt details. For couple tracking: individual partner data.



6.3 Word Document Export

Generates a comprehensive retirement plan report as an editable Word (.docx) document. The report covers your financial inputs and assumptions, year-by-year projections with embedded chart images (Portfolio Balance, Income vs Spending, Annual Spending Breakdown, and Partner Super in couple mode), a Portfolio Assessment & Recommendations section with plan-specific strengths, risks, and recommended actions, Monte Carlo risk analysis (if run), formal stress test results (if run), sensitivity analysis, and a methodology appendix explaining how the calculator works.



How to Generate

Configure your scenario and run whichever test scenarios you want included in the report — Monte Carlo is recommended for the risk analysis section, and formal stress tests are optional. Then click the "  Export Word" button at the top right of the screen. The document downloads automatically.

File Details

The file is a standard Microsoft Word .docx, typically 8–12 pages, named retirement-plan-YYYY-MM-DD.docx. It uses consistent professional formatting and is fully editable — you can add notes, highlight key figures, or customise the document before sharing.

When to Export

The Word export is most useful before major decisions (retiring, making large withdrawals, reviewing with a financial advisor), for periodic reviews to track how your plan is evolving, or when sharing with a spouse, partner, or estate planning professional. Export with different assumptions to build an offline set of scenario comparisons.

Differences from CSV Export

The Word report is a narrative document with summaries, risk assessments, and recommendations — suited to sharing and review. The CSV export contains raw year-by-year data for every calculated field — suited to custom analysis in Excel. For comprehensive planning, both are worth keeping: export the Word report for discussion and the CSV for independent verification.

PART 7: TECHNICAL REFERENCE

7.1 Annual Simulation Sequence

Each year processes in this order:

1. Increment ages and year counter
2. Check partner mortality (if death scenario active)
3. Apply dynamic guardrails (if enabled, from Year 2)
4. Calculate total spending: base + splurge + aged care + debt + one-off expenses
5. Calculate total income: pension + age pension + other net income + additional income streams + pre-retirement income
6. Apply minimum drawdown from Main Super (mandatory, age-based percentage)
7. Cover spending using withdrawal waterfall (Section 7.2)
8. Process debt payments (interest monthly, excess to principal)
9. Process RAD payment (aged care entry) or refund (exit)
10. Apply investment returns to portfolio buckets
11. Index spending and pensions to CPI
12. Record year data
13. Check termination: balance $\leq$ \$0 or target years reached

7.2 Withdrawal Hierarchy (Waterfall)

Funds are withdrawn in priority order to manage sequence risk and preserve growth potential:

1. Cash Account (tax-free, no penalties)
2. Sequencing Buffer (defensive, lower volatility)
3. Main Super (or individual partner supers in couple mode)

Minimum drawdown from Main Super applied first (mandatory), then additional withdrawals follow waterfall. Couple mode: proportional withdrawals from retired partners only.

7.3 Age Pension Calculation

The calculator models Age Pension using Centrelink's asset test, income test, and deeming rules. The lower result from the two tests determines your entitlement.

What's Modeled

Deeming Rules: Financial assets (super, cash, buffer, and assessable assets outside super) are deemed to earn income at Centrelink rates regardless of actual earnings:

- Single: 1.25% on first \$64,200, then 3.25% on balance
- Couple: 1.25% on first \$106,200, then 3.25% on balance
- Deeming thresholds and rates are indexed to CPI each year

Income Test - All Sources Counted:

- Private pensions and annuities (Defined Benefit/Superannuation Annuity)
- Deemed income from financial assets (calculated above)
- Other Net Income and Additional Income Streams (rental income, part-time work, dividends, overseas pensions, and other non-super investment income)
- Pre-retirement income (in couple mode if one partner still working)

Note: Account-based pension minimum drawdowns are not counted as income (correctly excluded per Centrelink rules).

Asset Test

Assessable assets = Main Super + Sequencing Buffer + Cash Account + Assessable Assets Outside Super (excludes principal residence). Non-super assets are entered in today's dollars and automatically indexed to inflation.

Homeowner Thresholds:

- Single: \$321,500 (full pension), \$722,000 (cutoff)
- Couple: \$481,500 (full pension), \$1,085,000 (cutoff)

Non-homeowner: Add \$252,000 to all thresholds

Taper: Pension reduces \$3 per fortnight (\$78/year) for each \$1,000 above threshold

Income Test

Income Free Area (indexed to CPI):

- Single: \$5,668/year
- Couple: \$9,880/year (combined)

Taper: Pension reduces 50c for every dollar of income above the free area

Example (Single, \$400,000 super, \$25,000 rental income, March 2026 rates):

- Deemed income: $(\$64,200 \times 0.0125) + (\$335,800 \times 0.0325) = \$11,514$

- Rental income: \$25,000

- Total testable income: \$36,514

- Excess over free area: $\$36,514 - \$5,668 = \$30,846$

- Reduction: $\$30,846 \times 50\% = \$15,423$

- Income test result: $\$31,224 - \$15,423 = \$15,801/\text{year}$

Final Calculation

Age Pension = $\text{Min}(\text{asset test result}, \text{income test result})$

Couple Mode:

- Both partners 67+: Uses couple rate (max ~\$47,071/year combined)

- Only one partner 67+: Uses single rate for eligible partner

- Neither 67+: No Age Pension

All thresholds, free areas, and maximum pension rates are automatically indexed to CPI each year.

Age Pension Calculation Details Panel

A collapsible panel in the results column (labelled “Age Pension Calculation Details”) shows how the Age Pension is calculated for each year of retirement. This panel only appears when the Age Pension is enabled and a simulation has been run. It provides full transparency into the means testing logic so you can verify the calculator’s output against Centrelink estimates.

The table displays the following columns for each year from age 67 onwards:

- Rate Type: Whether couple or single rates are being applied (changes automatically if a partner dies in couple tracking mode)

- Assessable Assets: Total assets counted for means testing (super + buffer + cash + non-super assets, all inflation-indexed)

- Asset Test Threshold and Asset Test Pension: The lower threshold and the pension amount resulting from the assets test

- Deemed Income: Income calculated via Centrelink deeming rates on total financial assets

- Private Pension + Other: All non-deemed income sources (defined benefit pension, Other Net Income, pre-retirement income)

- Total Testable Income, Income Free Area, and Income Test Pension: The income test inputs and resulting pension amount
- Binding Test: Which test (assets or income) produces the lower pension and therefore determines the payment. Highlighted in amber.
- Final Age Pension: The annual pension amount. Shows “FULL” in green when receiving the maximum rate. Red rows indicate nil pension (assets above the cutoff threshold).

All values in the table respect the Real/Nominal dollars toggle, so you can view the calculations in either today’s purchasing power or future nominal dollars.

Important Limitations

What’s Not Modelled:

- Work Bonus: Now modelled — see Section 2.4 for how to flag employment income streams as Work Bonus eligible.
- Pension Loan Scheme: Voluntary reverse mortgage option not modeled
- Energy Supplement: ~\$365/year supplement not added
- Commonwealth Seniors Health Card: Concessions not quantified

Verification Required: This is a planning tool. Actual Age Pension entitlement is determined by Centrelink based on your complete financial situation. Always verify estimates with Centrelink or a financial advisor, especially if you have:

- Rental income or investment income outside super
- Part-time work during retirement
- Complex asset structures (trusts, companies, etc.)
- Income from overseas sources

7.4 Minimum Drawdown Rules

Mandatory annual withdrawal percentages from Main Super:

- Age < 65: 4%
- Age 65-74: 5%
- Age 75-79: 6%
- Age 80-84: 7%
- Age 85-89: 9%

- Age 90-94: 11%
- Age 95+: 14%

Applied to Main Super balance at start of year. If minimum > available, withdraw full balance.
Couple mode: separate minimums for each partner based on individual age and balance.

7.5 Returns & Inflation Modeling

Constant Return slider represents NOMINAL returns (before inflation). Real return = Nominal - CPI.
Example: 7% nominal with 2.5% CPI = 4.5% real growth.

Main Super earns scenario-based returns (constant, historical, or Monte Carlo). Buffer and Cash earn the user-configured Cash/Buffer Return Rate (default 3% per year, set in Key Assumptions). This rate is applied nominally — it does not automatically add CPI, so a rate of 3% with 2.5% CPI implies approximately 0.5% real growth on defensive assets.

All spending and pensions indexed to CPI annually. Display mode (Real/Nominal) affects chart presentation only, not calculations.

APPENDIX: Version History

Version 17.2 (March 2026)

Sustainable Spending Finder: A “Find sustainable spending” panel added below the Base Annual Spending field. Two modes — Constant Return (binary search finding the maximum spending that hits a target ending balance at the Modelled Death Age) and Monte Carlo (binary search finding the spending level that achieves a chosen success rate threshold: 70–90%). Both modes use the full simulation engine including Age Pension means testing, couple tracking, guardrails, and all income sources. Splurge spending is included on top of the base figure in every iteration. An Apply button populates the Base Annual Spending field directly.

Word export enhanced: embedded chart images (Portfolio Balance, Income vs Spending, Annual Spending Breakdown, Partner Super) now appear in the Detailed Results section; new Portfolio Assessment & Recommendations section generates plan-specific strengths, risks, and recommended actions from formal test results and MC success rates.

Version 17.1 (March 2026)

Variable simulation horizon: The simulation now runs to the Modelled Death Age rather than a fixed 35-year window. In single mode, the death age is set via a new field (defaulting to ABS 2020–2022 life expectancy for the selected sex and age) and the simulation horizon adjusts automatically. In couple tracking mode, a separate Simulation Length (years) field controls how long the scenario runs, completely independent of the partner death ages used in Death Scenario Modelling.

Stochastic mortality in Monte Carlo: A “Variable death age in MC” checkbox enables per-run mortality sampling. Each parametric MC run draws a death age from a normal distribution centred on the ABS life expectancy (SD = 10 years), so run lengths vary across the ensemble and the success rate reflects genuine longevity uncertainty rather than a fixed horizon. Historical MC continues to use fixed 35-year windows to preserve the integrity of historical return sequences.

Longevity & Outcome Analysis chart: A new chart appears below the Executive Summary when Monte Carlo is active, providing two views — a scatter plot of final balance vs end age (green = solvent, red = insolvent) that reveals whether failures are longevity-driven or return-driven, and a ruin age histogram showing the distribution of portfolio depletion ages for failed runs.

Age Pension parameters updated to 20 March 2026 indexation: single maximum \$31,224/yr (up from \$30,646), couple combined \$47,071/yr (up from \$46,202). Single homeowner asset cutoff increased to \$722,000 (up \$7,500). Deeming rates increased to 1.25% / 3.25% (up 0.50% each). Lower asset thresholds and income free areas unchanged until 1 July 2026.

Death scenario survivor pension fix: The surviving partner now correctly receives the full single Age Pension rate after their partner dies, rather than the lower member-of-couple rate that was incorrectly applied in prior versions.

Sensitivity Analysis hidden in MC and Formal Test modes: The panel only appears for Constant Return and Historical Period scenarios, where a deterministic return grid is meaningful.

Version 17.0 (March 2026)

Updated Parametric Monte Carlo: replaced norm arithmetic normal return modelling with a lognormal return framework to better represent portfolio behavior and volatility drag.

Version 16.0 (February 2026)

Work Bonus (Age Pension Employment Income Exemption): Employment and self-employment income can now be flagged as Work Bonus eligible on a per-stream basis. Tick the “Employment income — Work Bonus eligible” checkbox beneath the main Other Net Income field or on any Additional Income Stream card to exempt up to \$7,800/year (per eligible partner) from the Age Pension income test. Unused entitlement accumulates in a bank up to \$11,800 per person. Both amounts are CPI-indexed. The bank starts at \$0. Rental income, dividends, and pensions are not eligible and remain fully assessable.

Australian Historical Return Data: Three Australian data series added to Historical Monte Carlo: AU Balanced (60% XAOA / 40% AusBond Composite, 1990–2024, 35 years), AU Equities — XAOA ASX All Ords Accumulation (1980–2024, 45 years), and AU Bonds — AusBond Composite 0+ Yr (1990–2024, 35 years). These complement the existing US S&P 500 dataset (1928–2025, 98 years). AU Balanced is now the default series, providing a more realistic starting point for typical Australian super fund members.

Asset Allocation Presets (Parametric Monte Carlo): Four one-click presets replace the need to manually enter return and volatility assumptions: Conservative (~30% equities, 6.5%/8%), Balanced (60/40, 7.5%/11%, new default), Growth (70/30, 8.0%/13.5%), and High Growth (~90% equities, 9.0%/18%). A Custom option retains full manual control. Preset names and descriptions appear in the Word export Analysis Configuration and Assumptions sections.

Word Export Enhancements: The Word report now reflects the actual selected data series, preset name, and Work Bonus configuration. The Analysis Configuration table shows the correct series label and period (e.g., “AU Balanced (60% XAOA / 40% AusBond Composite) 1990–2024”) rather than the hardcoded S&P 500 description. The Assumptions section lists the preset name alongside return and volatility figures. Work Bonus eligibility for each income stream is shown in the Other Income table. A new Work Bonus sub-section in the Methodology Appendix explains the accrual mechanics and eligibility rules. Historical Period results now appear correctly in the Word report regardless of the order in which test scenarios are selected.

Historical Periods — Australian Data: The Historical Periods test scenario now supports four data sources: US S&P 500 (unchanged, 1928–2025), AU Equities — XAOA (1980–2024), AU Balanced 60/40 (1990–2024), and AU Bonds — AusBond Composite (1990–2024). Eleven periods are available in total (6 US + 5 AU). AU periods use event-driven labels (e.g., “AU 1990 Recession”, “AU 1987 Crash”) to make clear which are genuine stress tests and which are best-case comparators. A contextual description beneath the selector explains which events each 35-year sequence captures.

Historical Periods — Data Integrity and [Stress Test] Labels: Three of the six US S&P 500 periods (2008 GFC, 2000 Dot-com, 2020 COVID) cannot provide a full 35 years of real data because

the events occurred too recently — a 2008 start requires data through to 2043, which does not yet exist. These periods are now clearly labelled [Stress Test] in the selector to distinguish them from the three genuine 35-year replays (1929 Depression, 1973 Stagflation, 1982 Bull Market). The fabricated tail years in the stress test periods previously used recycled values averaging approximately 18%/yr, which was materially optimistic and obscured the true impact of the opening crash. These have been replaced with a flat 10%/yr nominal return — the long-run S&P 500 historical average — as a neutral, defensible assumption. Each period's description panel now states exactly how many years of real data are used and what assumption applies to the remainder. All AU periods use genuine historical data throughout and are unaffected by this change.

Version 15.9 (February 2026)

Non-Super Assets for Age Pension: A new “Assessable Assets Outside Super” field allows you to enter financial assets held outside superannuation (bank accounts, term deposits, shares, managed funds, investment property equity). These are added to your super balance for both the Centrelink assets test and deeming calculation, providing a more accurate Age Pension estimate. The field appears when Age Pension is enabled and values are automatically indexed to inflation. Income from these assets (rent, dividends) should be entered in the existing “Other Net Income” field.

Age Pension Calculation Details Panel: A new collapsible panel in the results column shows a year-by-year breakdown of how your Age Pension is calculated. For each year from age 67 onwards, it displays: total assessable assets, asset test threshold and result, deemed income, total testable income, income test free area and result, which test is binding (assets or income), and the final pension amount. Values respect the Real/Nominal dollars toggle. Rows are colour-coded: amber highlights the binding test, green marks full pension rate, and red indicates nil pension (above asset cutoff).

Security & Analytics: Added Vercel Analytics for privacy-friendly visitor tracking. Content Security Policy (CSP) headers and other security hardening measures protect against common web vulnerabilities.

Additional Income Streams: You can now add multiple income sources, each with its own description, amount, start year, duration, and CPI indexing setting. This solves a key limitation where users with both indexed and non-indexed income (e.g., a CPI-indexed rental income and a fixed overseas pension) could not model them correctly with a single field. Each stream is independently calculated and flows into the income charts, spending waterfall, and Age Pension income test. The original “Other Net Income” field remains for backward compatibility; additional streams appear below it via the “Add Income Stream” button.

What-If Panel Cleanup: The “Run Comprehensive Analysis” and “Save Current Scenario” buttons have been moved inside the What-If Comparison panel, reducing visual clutter when the panel is collapsed.

Version 15.8 (February 2026)

Recalibrates the JP Morgan spending curve to align with Blanchett (2014) and JP Morgan (2024) research, adjusted for Australia: a gentler 1.0%/yr decline in the go-go years, accelerating to

1.5%/yr in the slow-go years, then tapering to 0.5%/yr in the no-go years. Stochastic irregular expenses are now enabled by default (with pre-loaded manual expenses removed to avoid double-counting) and can run simultaneously with manual one-off expenses for planned costs like gifts or holidays. A new 🎲 Reshuffle Expense Timing button lets you explore different stochastic expense paths without running full Monte Carlo. Several simulation stability fixes resolve issues where charts could jump around during editing (unseeded random values in aged care probability and the stochastic expense engine), the Monte Carlo stale detection "orange button" warning now works as documented, and debt repayment changes correctly update the spending charts. Chart tooltips now display age alongside year.

Version 15.7 (February 2026)

Two-Column Layout: Calculator now uses a side-by-side layout on desktop screens. Input parameters appear in the left column (approximately 33% width), while test scenarios, charts, and results appear in the right column (approximately 67% width). This allows you to see inputs and results simultaneously without scrolling back and forth. Test scenario buttons are now positioned at the top of the right column for easy access. The single-row button arrangement at the top (Save/Load, Quick Help, Tax Assumptions, Key Assumptions, Export CSV, Export Word) reduces vertical space and keeps all controls visible. On mobile devices and tablets, the layout automatically stacks vertically for optimal viewing.

Version 15.6 (February 2026)

Annual Spending Breakdown Chart: New stacked bar chart shows the composition of your annual spending by funding source. Each bar displays how much comes from Super withdrawals (blue), Cash (yellow), Sequencing Buffer (orange), Age Pension (dark green), Defined Benefit pension (medium green), and Other Income (light green). In single-scenario mode, a minimum required super withdrawal line overlays the chart, making it easy to verify whether your super withdrawals meet the legal minimum. This visualization reveals when Age Pension kicks in, shows the impact of large one-time expenses like RAD payments, and helps assess the sustainability of your drawdown strategy.

Age Pension Improvements: Comprehensive Age Pension modeling now includes Centrelink deeming rules (financial assets deemed to earn 0.25% on first portion, 2.25% on balance) and complete income testing (counts private pensions, deemed income, Other Net Income from rental/work, and pre-retirement income). Added clear warnings about modeling assumptions and verification with Centrelink.

Historical Monte Carlo Success Rates: Updated success rate guidance to reflect that Historical MC typically shows 5-15 percentage points higher than Parametric MC. Historical MC "Good" threshold raised to 85-89% (from 80-89%) to reflect that it's bounded by actual historical data. Added comparative explanation panel when both methods are run.

Other Net Income: Model additional income sources (part-time work, rental properties, investments) with flexible start year and duration controls. Works in both single and couple modes with year-based timing.

Couple Mode Fixes: Real dollar year now displays correctly using earliest retirement age. Retirement year shown next to age in couple tracking panel.

Version 15.5 (February 2026)

Terminology Updates: Renamed “PSS/CSS Pension” to “Defined Benefit/Superannuation Annuity” throughout to be more inclusive of all pension types (PSS, CSS, DFRDB, commercial lifetime annuities). Renamed “Base Annual Spending” to “Base Annual Spending in Retirement” for clarity. Renamed “Current Age” to “Age turning this year” to avoid confusion with birthdays during the year. Updated “Withdrawal remainder” to “Withdrawal order” in strategy description for better clarity.

Word Document Export: Generate comprehensive retirement plan reports as professional Word documents. Includes executive summary with traffic light status indicators, financial assumptions, year-by-year projections (constant return scenario), Monte Carlo risk analysis, actionable recommendations with priority coding, and detailed test scenario results. Reports are fully editable, 8-12 pages, with professional styling. Export button located next to CSV Export for easy access.

Version 15.4 (February 2026)

Save/Load Scenarios: Save multiple retirement scenarios to browser storage with custom names. Load previously saved scenarios instantly. Export scenarios as JSON files for backup or sharing. Import scenarios from JSON files. Auto-save draft every 30 seconds with recovery on page load.

Stochastic Irregular Expenses Enhanced: Now works with all test scenarios (Constant Returns, Historical Periods, Monte Carlo, Historical Monte Carlo). Previously only worked with Monte Carlo. Generates realistic lumpy expenses (vehicles, home repairs, medical) for single-run scenarios with proper inflation calculation from retirement year.

Enhanced Warning System: Comprehensive early depletion warning when portfolio fails in first 5 years. Identifies root causes (low returns, high spending, stochastic expense spikes, couple retirement gaps). Provides specific actionable fixes including pre-retirement income guidance for staggered retirement scenarios.

Manual Expense Visibility Improved: Manual one-off expenses now always visible even when stochastic is enabled, with clear warnings about which system is active. Orange warning when using manual expenses with Monte Carlo explains limitation (same expenses in every simulation).

Stale Results Detection: Monte Carlo and Historical Monte Carlo buttons turn orange and pulse when settings change after running. Clear "🔄 Re-Run" prompts with warning messages ensure users know when results are outdated.

Streamlined Interface: Removed Quick Start Templates from Scenario Manager to reduce clutter. Focus on user-created scenarios with enhanced save/load functionality.

Version 15.2 (February 2026)

Executive Summary Dashboard: Collapsible panel above charts showing success/fail status, portfolio metrics, income summary, risk assessment with traffic lights (● ● ●), Monte Carlo analysis, and couple tracking. Scenario-aware with hover tooltips on all metrics.

Version 15.1 (February 2026)

Monte Carlo Percentile Bands: Visualize outcome uncertainty with 10th, 25th, 75th, 90th percentile boundary lines.

Individual Partner Charts: Track each partner's super balance and pension income separately with couple tracking.

Accurate Age Pension: Implements real Centrelink rules for couples with different ages and eligibility.

Version 15.0 (February 2026)

Couple Tracking: Complete individual partner management with different ages, retirement dates, super balances, and pensions. Includes death scenarios and reversionary pension modeling.

Contact & Support

Email: aust-retirement-calculator@proton.me

Documentation: Available in calculator (Help & Guide section)

User Guide: Download PDF from calculator interface

Important: This tool is for educational and planning purposes only. It does not provide financial, tax, or retirement advice and does not consider your personal circumstances. Consult qualified professionals for personalized advice.

End of User Guide